

One-Shot Generative Flows: Existence and Obstructions

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Abstract

We study dynamic measure transport in generative modelling via stochastic process $X_\bullet$ whose marginals interpolate between a source distribution P_0 and a target distribution P_1 while remaining independent, i.e., when $(X_0, X_1) \sim P_0 \otimes P_1$. The conditional expectation of the process velocity defines an ODE whose flow map transports from P_0 to P_1 . We discuss when such a process induces a *straight-line flow*, namely one whose pointwise acceleration vanishes and is therefore exactly integrable by any first-order method. We first develop multiple characterizations of straightness in terms of PDEs involving the conditional statistics of the process. Then, we prove that straightness under endpoint independence exhibits a sharp dichotomy. On one hand, we construct explicit, computable straight-line processes for arbitrary Gaussian endpoints. On the other hand, we show straight-line processes do not exist for targets with sufficiently well-separated modes. We demonstrate this through a sequence of increasingly general impossibility theorems that uncover a fundamental relationship between the sample-path behavior of a process with independent endpoints and the space-time geometry of this process' flow map. Taken together, these results provide a structural theory of when straight generative flows can, and cannot, exist.

1 Introduction

Sampling from complex probability distributions is central to probabilistic inference and modern generative modelling. A recent line of work establishes *dynamic measure transport* as a unifying paradigm: First, construct a stochastic process $(X_t)_{t \in [0,1]}$ designed to have marginals that interpolate a source distribution P_0 at $t = 0$ and a target distribution P_1 at $t = 1$. Then, estimate the *conditional velocity*

$$v_t(x) := \mathbb{E}[\dot{X}_t \mid X_t = x]$$

and generate samples by evaluating the ODE flow maps ϕ_t defined by

$$\partial_t \phi_t(x) = v_t(\phi_t(x)) \quad \text{with} \quad \phi_0(x) = x.$$

This perspective underlies methods such as *stochastic interpolants* [ABVE23, AVE22], *flow matching* [LCBH⁺23, TFM⁺24a], and *score-based probability flow ODEs* [SSDK⁺21], as well as *rectified*

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flows [LGL22, Liu22, BRSR25, HCD25]—all of which have demonstrated strong empirical performance.

The computational efficiency of these methods hinges on the *geometry of the induced flow*. Generic flows demand many velocity-oracle evaluations because numerical integration error scales with the curvature (and higher derivatives) of ϕ_t . In contrast, if the flow is *straight*, meaning that the acceleration of the flow map vanishes,

$$\partial_{tt} \phi_t(x) \equiv 0 \quad \text{for all } (x, t),$$

then $\phi_t(x)$ is affine in t : $\phi_t(x) = (1-t)x + t\phi_1(x)$. Consequently, any first-order integrator is exact; one can traverse the entire path with a single velocity evaluation. This motivates a fundamental question:

Which stochastic processes $(X_t)_t$ with $X_0 \sim P_0$ and $X_1 \sim P_1$ induce straight flows?

Prior work (e.g., [LGL22, Liu22, BRSR25]) offers algorithmic frameworks that convert a given non-straight flow into a straight one, but a *structural characterization* of when straightness is intrinsic to the underlying process has remained open. This paper develops such a theory.

One way to trivially obtain a straight flow is via an optimal transport coupling, see [McC97]. Namely, let $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be such that $T_{\#}P_0 = P_1$ minimizing the Monge cost and set

$$X_t = (1-t)X_0 + tT(X_0) \quad \text{with } X_0 \sim P_0, \quad (1)$$

assuming that $T = \nabla\psi$ is the gradient of a strictly convex potential $\psi : \mathbb{R}^d \rightarrow \mathbb{R}$. It is not hard to see that the resulting flow is straight, owing to the fact that each sample path $t \mapsto X_t(\omega)$ is straight and no two sample paths intersect. In fact, any transport map T with positive definite Jacobian $\nabla T(x)$ at each $x \in \mathbb{R}^d$ yields such a trivial solution, see Theorem 5 for details—examples include Monge maps corresponding to strictly convex potentials and Knöthe–Rosenblatt rearrangements between densities with minimal regularity. Such a solution is circular, however, as it constructs a process $X_{\bullet}$ to sample P_1 in terms of a transport map T . If we had access to T , we could sample P_1 by sampling P_0 and applying T without ever constructing $X_{\bullet}$.

Recent literature has used variations of this idea successfully to build sampling algorithms, essentially by constructing near optimal couplings that yield nearly straight flows. Indeed, rectified-flow methods, minibatch-coupling methods, and optimal-transport variants of conditional flow matching exploit this fundamental property of optimal transport, yielding practically relevant algorithms, see [LGL22, PBHDE⁺23, TFM⁺24b]. Such algorithms, however, require additional up-front computation: first estimate some surrogate $\tilde{T}$ to a transport map T and then instantiate a simple process $X_{\bullet}$ in terms of $\tilde{T}$. A practical question is whether one can bypass this: can we obtain straight-flows by a well-crafted $X_{\bullet}$ without any transport map computation? Perhaps more fundamentally: are straight line flows a unique property to the processes (1) relying on a transport map T with a positive definite Jacobian?

To explore this rigorously, we impose the *independent endpoint constraint*, that is, we ask that the candidate process $X_{\bullet}$ satisfies

$$(X_0, X_1) \sim P_0 \otimes P_1.$$

This assumption is natural as typically we have a data set consisting of i.i.d. samples from P_1 and we wish to construct the process $X_{\bullet}$ by drawing i.i.d. samples from P_0 and connecting pairs

(X_0, X_1) . In this setting, there is no relationship between source and target samples. The only coupling we can impose a priori, i.e., without any computation, is the product measure $P_0 \otimes P_1$. Thus, our final formulation and the central question of this paper is:

Given P_0 and P_1 , which stochastic processes $(X_t)_t$ with $(X_0, X_1) \sim P_0 \otimes P_1$ induce straight flows?

To study this problem, we introduce two classes of stochastic processes. The first, $\mathcal{S}_{\text{SL}}(P_0, P_1)$, consists of independent endpoint processes with suitable regularity that induce straight-line flows. The second, $\mathcal{F}_{\text{SL}}(P_0, P_1)$, consists of those members of $\mathcal{S}_{\text{SL}}(P_0, P_1)$ that admit a computationally tractable representation—we codify this by introducing *generalized interpolants*, a mild generalization of the stochastic interpolants framework [\[ABVE23\]](#). This distinction lets us separate structural possibility from constructive realizability.

1.1 Outline

We begin with an analytical reformulation of straightness. Straight-line flows can be characterized in terms of both the conditional velocity of a stochastic process as well as the associated flow

$$X \in \mathcal{S}_{\text{SL}}(P_0, P_1) \iff \partial_{tt}\phi_t = 0 \iff D_t v_t = 0,$$

where $D_t v_t = \partial_t v_t + (v_t \cdot \nabla) v_t$. We also show the equivalence of the above statements with a balance law

$$\nabla \cdot (\rho_t \Pi_t) = \rho_t a_t,$$

with the marginal density ρ_t , ensemble acceleration $a_t(x) = \mathbb{E}[\ddot{X}_t \mid X_t = x]$, and Π_t as a Reynolds-type conditional covariance tensor. This reformulation gives insight to the most classical setting, demonstrating that *affine processes*

$$X_t = (1 - t)X_0 + tX_1$$

can induce straight-line flows only if the coupling of (X_0, X_1) is deterministic, i.e., there exists a transport map T such that $X_1 = T(X_0)$ almost surely—this is also sufficient, provided the transport map satisfies certain properties. This formalizes existing intuition that deterministic coupling is necessary for straightness, given a simple process design.

Despite this result and the independence constraint, straight-line generalized interpolants do exist for arbitrary Gaussian endpoints. We construct explicit processes of the form

$$X_t = (1 - t)X_0 + tX_1 + \sqrt{2t(1 - t)} Z,$$

with $(X_0, X_1, Z) \sim P_0 \otimes P_1 \otimes Q$, where Q is a carefully selected Gaussian measure. The construction is reminiscent of Brownian- and Schrödinger-bridge corrections, but here the auxiliary noise is tuned to enforce vanishing material derivative rather than entropic optimality [\[BTHD21\]](#) [\[TMF⁺24\]](#).

Our third contribution reveals a fundamental structural obstruction. We show that, for multi-modal targets with sufficiently separated modes, straight-line flows and endpoint independence are fundamentally incompatible. The intuition is geometric: straight-line characteristics impose rigid space-time geometries on the flow ϕ which cannot be created by independent-endpoint processes. We demonstrate this with three impossibility theorems of increasing generality and sophistication: First,

we argue the case of measures with disjoint support in dimension $d = 1$ using purely topological arguments, which distills the essence of the obstruction. Next, we extend this to measures with disjoint support in $d > 1$. Finally, we return to $d = 1$ and treat measures with possibly full support on $\mathbb{R}$ but appropriately concentrated “multi-modal” profiles; here, we need to develop sophisticated analytical arguments and make assumptions on the sample-path regularity of the process. The resulting picture is conceptually sharp. Independent endpoints do not make straight-line generation impossible in general, as Gaussian targets admit explicit, computable, one-shot constructions. But neither is straightness universal, since multi-modality possesses a fundamental obstruction. Thus, we show a dichotomy in the design and capabilities of modern samplers based on stochastic processes, uncovering fundamental properties of such generative flows.

1.2 Contributions

Our main contributions are as follows:

1. *Analytical reformulation and characterization of straight-line processes.* We develop a new analytical framework for straightness. In particular, we show that for a broad class of stochastic processes, straight-line generation can be characterized equivalently in terms of no curvature of the induced flow map, the vanishing of the material derivative of the conditional velocity, and a new balance law involving the marginal density, ensemble acceleration, and a Reynolds-type conditional covariance tensor. As a first consequence, we completely characterize the classical affine processes

$$X_t = (1 - t)X_0 + tX_1,$$

proving that they induce straight flows if and only if the endpoint coupling is deterministic.

2. *Existence of straight-line processes under endpoint independence.* We show that the independent endpoint constraint does not preclude straight-line generation in general. For arbitrary Gaussian marginals, we construct explicit, computable processes with

$$(X_0, X_1) \sim P_0 \otimes P_1,$$

which yield exact straightness. We provide visualizations in Section [1.5](#).

3. *Obstructions for multi-modal targets.* We prove that straightness under endpoint independence fails in a broad class of multi-modal settings. Starting from $d = 1$ -dimensional measures with disjoint support, we then progress to measures in $d > 1$ with disjoint support as well as measures in $d = 1$ with full support. In doing so, we demonstrate that strong modal separation is fundamentally incompatible with straight-line generation from independent endpoints.

Scope and implications. The results in this manuscript are largely theoretical, targeting straight-line generation at the level of general stochastic processes rather than a specific algorithmic parameterization. Our results yield a structural picture with both positive and negative sides: straight-line generation under endpoint independence is possible for some classes of targets, but impossible for others. We view this as a first step toward a principled theory for the design of generative flows.

1.3 Notation

Below we review notation used in this work.

1.3.1 Probability Theory

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be an abstract probability space. Random variables are measurable functions $X : \Omega \rightarrow \mathbb{R}^d$, ¹ and the expectation $\mathbb{E}[X]$ is defined as the integral $\int_{\Omega} X d\mathbb{P}$. For random variables $X, Y : \Omega \rightarrow \mathbb{R}^d$ the conditional expectation $\mathbb{E}[X | Y]$ is defined as the conditional expectation $\mathbb{E}[X | \sigma(Y)]$ where $\sigma(Y)$ is the σ -algebra generated by Y and for $y \in \mathbb{R}$ the conditional expectation $\mathbb{E}[X | Y = y]$ is the integral $\int_{\Omega} X d\mathbb{P}_y$ and $\mathbb{P}_y$ is the *disintegration* [Bog07, Section 10.6] of $\mathbb{P}$ on the level sets of Y . Finally, given another measurable space $(\Omega', \mathcal{F}')$ we denote the set of measures on Ω' by $\mathcal{P}(\Omega')$ and the subset of absolutely continuous measures by $\mathcal{P}_{\text{a.c.}}(\Omega')$.

1.3.2 Analysis

Let $C^k([0, 1]; \mathbb{R}^d)$ be the space of component-wise k -times continuously differentiable functions from $[0, 1]$ to $\mathbb{R}^d$ and $W^{k,p}([0, 1]; \mathbb{R}^d)$ to be the space of component-wise k -times weakly differentiable functions $[0, 1] \rightarrow \mathbb{R}^d$ with weak (component-wise) derivatives in $L^p([0, 1])$. An $\mathbb{R}^d$ -valued stochastic process is a collection $\{X_t\}_{t \in I}$ indexed by some set I , where $X : \Omega \rightarrow \mathbb{R}^d$ is measurable. In this paper, we take $I = [0, 1]$ and write $X := (X_t)_t := (X_t)_{t \in [0, 1]}$. A sample path of a stochastic process is the function $t \mapsto X_t(\omega)$ for a fixed realization $\omega \in \Omega$. We note that a stochastic process with sample paths in the Sobolev space $W^{k,p} := W^{k,p}([0, 1]; \mathbb{R}^d)$ can equivalently be viewed as a random variable $X : \Omega \rightarrow W^{k,p}$ and its law is thus in $\mathcal{P}(W^{k,p})$.

For real numbers $a, b \in \mathbb{R}$ we denote their minimum and maximum by

$$a \wedge b := \min(a, b), \quad a \vee b := \max(a, b).$$

For a set $S \subseteq \mathbb{R}^d$ its topological closure is denoted by $\bar{S}$ and it is defined as the smallest closed set containing S , namely

$$\bar{S} := \bigcap_{\substack{S \subseteq C \subseteq \mathbb{R}^d \\ C \text{ closed}}} C.$$

The interior of S is denoted by S° and it is defined as the largest open set contained in S , namely

$$S^\circ := \bigcup_{\substack{C \subseteq S \subseteq \mathbb{R}^d \\ C \text{ open}}} C,$$

and the boundary of S is defined as

$$\partial S := \bar{S} \setminus S^\circ.$$

The convex hull of S is denoted by $\text{conv } S$ and it is defined as the smallest convex set containing S , namely

$$\text{conv } S := \bigcap_{\substack{S \subseteq C \subseteq \mathbb{R}^d \\ C \text{ convex}}} C.$$

The closed convex hull of S is denoted by $\overline{\text{conv}} S$ and it is defined as the smallest closed convex set containing S , namely

$$\overline{\text{conv}} S := \bigcap_{\substack{S \subseteq C \subseteq \mathbb{R}^d \\ C \text{ closed and convex}}} C.$$

¹In this paper, subsets of $\mathbb{R}^d$ are always equipped with the Borel σ -algebra.

For a finite subset $S = \{x_1, \dots, x_n\}$ of $\mathbb{R}^d$ we write $|S|$ for the cardinality of S . For an infinite subset $S \subseteq \mathbb{R}^d$ we write $|S|$ for the Lebesgue measure of S . The distinction will be clear from context. For any two sets $A, B \subseteq \mathbb{R}^d$ we define their distance as

$$\|A - B\| := \inf_{a \in A, b \in B} \|a - b\|,$$

and $\|\cdot\|$ is the Euclidean norm on $\mathbb{R}^d$. For $x \in \mathbb{R}^d$ if the set $A = \{x\}$ is a singleton we mildly abuse notation and write $\|x - B\|$ to mean

$$\|x - B\| := \inf_{b \in B} \|x - b\|.$$

1.3.3 Dynamics & Linear Algebra

Fix a stochastic process $X := (X_t)_{t \in [0,1]}$ with sample paths in the Sobolev space $W^{2,2}([0,1]; \mathbb{R}^d)$. We define the *conditional velocity* and *conditional acceleration* fields, also referred to as the *ensemble velocity* and *ensemble acceleration*, by

$$v(t, x) := \mathbb{E} \left[\dot{X}_t \mid X_t = x \right] \quad \text{and} \quad a(t, x) := \mathbb{E} \left[\ddot{X}_t \mid X_t = x \right].$$

We define the *second moment velocity tensor* and the *covariance* or *Reynolds stress tensor* as

$$\mathcal{C}(t, x) := \mathbb{E} \left[\dot{X}_t \otimes \dot{X}_t \mid X_t = x \right] \quad \text{and} \quad \Pi(t, x) := \mathcal{C}(t, x) - v(t, x) \otimes v(t, x),$$

respectively. We let $\mu_t = \text{Law}(X_t)$ be the marginal law of the process X at time t and write ρ_t for the density of μ_t with respect to the Lebesgue measure, if it exists.

We define the *material derivative* of velocity v at $x \in \mathbb{R}^d$ as

$$D_t v(t, x) := \partial_t v(t, x) + (v(t, x) \cdot \nabla) v(t, x).$$

Because we often consider the space $\mathbb{R}^d$ over the time domain $[0, 1]$, we often denote time dependent velocity vector field as v_t defined as the mapping $x \mapsto v(t, x)$, with similar definitions for a_t , $\mathcal{C}_t$, Π_t . Finally, for matrices $A, B \in \mathbb{R}^{d_1 \times d_2}$ we denote the *Frobenius inner product* by

$$A : B := \text{Tr}(A^\top B) = \sum_{i=1}^{d_1} \sum_{j=1}^{d_2} A_{ij} B_{ij},$$

and for a matrix field $V : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ consisting of differentiable entries we define its *divergence* as the vector field $\nabla \cdot V : \mathbb{R}^d \rightarrow \mathbb{R}^d$ with components $(\nabla \cdot V)_j(x) = \sum_{i=1}^d \partial_i V_{ij}(x)$. We also denote the space of symmetric positive semi-definite matrices of dimension $d \times d$ by $\mathbb{R}_+^{d \times d}$ and that of strictly positive definite matrices by $\mathbb{R}_{++}^{d \times d}$.

1.4 Preliminaries

Now we introduce the main objects of interest in this work, namely the class of *stochastic processes with continuous sample paths* and the class of *generalized interpolants*. Let us remark, here, on the meaning of these objects: most applications in probabilistic generative modelling rely on relatively simply stochastic processes, e.g. affine processes seen above, Ornstein–Uhlenbeck processes,

Brownian bridges etc. Our goal in this work is *design*, as we seek to understand what processes, if any, could yield desirable behavior of these models. As such, we are led to two sub-goals: First, if non-trivial processes with some desirable behavior exist, describe them in a form amenable to computation. This is the role played by generalized interpolants, which are processes that could easily be instantiated on a computer. Our second goal is to characterize obstructions to computational desiderata by proving impossibility, i.e. “no-go theorems”, for the largest class of processes relevant for generative modelling. That is the role played by the class of stochastic processes with continuous sample paths.

We will now define preliminary notions that will lead us to the main objects of interest.

Definition 1. An $\mathbb{R}^d$ -valued stochastic process $X_\bullet := (X_t)_{t \in [0,1]}$ is said to have *continuous sample paths* if for $\mathbb{P}$ -a.s. $\omega \in \Omega$ the function $t \mapsto X_t(\omega)$ is continuous. Given $\mathbb{R}^d$ -measures P_0 and P_1 we write $\mathcal{S}(P_0, P_1)$ for the class of all $\mathbb{R}^d$ -valued stochastic processes with continuous sample paths satisfying

$$(X_0, X_1) \sim P_0 \otimes P_1.$$

Definition 2 (Base interpolation path). Fix a $1 \leq p < 2$. A *base interpolation path* is a Borel measurable mapping

$$\begin{aligned} F : [0, 1] \times \mathbb{R}^{2d+\ell} &\rightarrow \mathbb{R}^d \\ (t, x, y, z) &\mapsto F(t, x, y, z) \end{aligned}$$

such that $t \mapsto F(t, x, y, z)$ is in $W^{1,p}([0, 1]; \mathbb{R}^d)$ for (x, y, z) -Lebesgue a.s. in $\mathbb{R}^{2d+\ell}$ while also satisfying the boundary conditions

$$\begin{aligned} F(0, x, y, z) &= x \\ F(1, x, y, z) &= y \end{aligned}$$

for (x, y, z) -Lebesgue a.s. in $\mathbb{R}^{2d+\ell}$. The set of all such mappings is denoted by $\mathcal{F}$.

Definition 3 (Generalized Interpolant). A *generalized interpolant* is a pair (F, Q) with $F \in \mathcal{F}$ and $Q \in \mathcal{P}_{\text{a.c.}}(\mathbb{R}^\ell)$ for some $\ell \in \mathbb{N}$. For measures $P_0, P_1 \in \mathcal{P}(\mathbb{R}^d)$ the induced stochastic process of (F, Q) is $X_\bullet : t \mapsto X_t$ defined via

$$X_t = F(t, X_0, X_1, Z)$$

for $(X_0, X_1, Z) \sim P_0 \otimes P_1 \otimes Q$.

Definition 4 (Straight-Line Processes). Fix absolutely continuous measures $P_0, P_1 \in \mathcal{P}(\mathbb{R}^d)$. A straight-line process connecting P_0 and P_1 is a stochastic process $X_\bullet \in \mathcal{S}(P_0, P_1)$ such that the conditional velocity $v(t, x) = \mathbb{E}[\dot{X}_t | X_t = x]$ is well-defined, measurable in the pair (t, x) , it satisfies

$$\|v(t, x) - v(t, y)\| \leq L \|x - y\| \tag{2}$$

for some $L > 0$ and all $(t, x, y) \in [0, 1] \times \mathbb{R}^d \times \mathbb{R}^d$ and the flow map $\phi(t, x)$ defined by

$$\begin{cases} \partial_t \phi(t, x) = v(t, \phi(t, x)), \\ \phi(0, x) = x, \end{cases}$$

satisfies

$$\partial_t^2 \phi(t, x) = 0 \quad \text{for all } (t, x) \in [0, 1] \times \mathbb{R}^d. \quad (3)$$

The class of all such processes is denoted $\mathcal{S}_{\text{SL}}(P_0, P_1)$, i.e.,

$$\mathcal{S}_{\text{SL}}(P_0, P_1) := \left\{ X_{\bullet} \in \mathcal{S}(P_0, P_1) : X_{\bullet} \text{ satisfies } \textcircled{2} \text{ and } \textcircled{3} \right\}. \quad (4)$$

The class of all generalized interpolants inducing straight-line processes connecting P_0 and P_1 is denoted by

$$\mathcal{F}_{\text{SL}}(P_0, P_1) := \left\{ (F, Q) \in \mathcal{F} \times \mathcal{P}_{\text{a.c.}}(\mathbb{R}^{\ell}) : X_{\bullet} \in \mathcal{S}_{\text{SL}}(P_0, P_1) \right\}. \quad (5)$$

Remark 1. Notice that even though a generalized interpolant (F, Q) is defined irrespective of the choice of marginals $P_0, P_1 \in \mathcal{P}(\mathbb{R}^d)$, the induced stochastic process $X_{\bullet}$ as well as the associated statistics v and ϕ do depend on P_0 and P_1 . If necessary, we will make this dependence explicit by writing $X_{\bullet}^{P_0, P_1}$, v^{P_0, P_1} and ϕ^{P_0, P_1} .

1.5 Visualizations

Below we visualize examples of processes in $\mathcal{F}_{\text{SL}}(P_0, P_1)$ for $d = 1, 2, 3$ and Gaussian measures P_0, P_1 . See Section [3.1](#) for details.

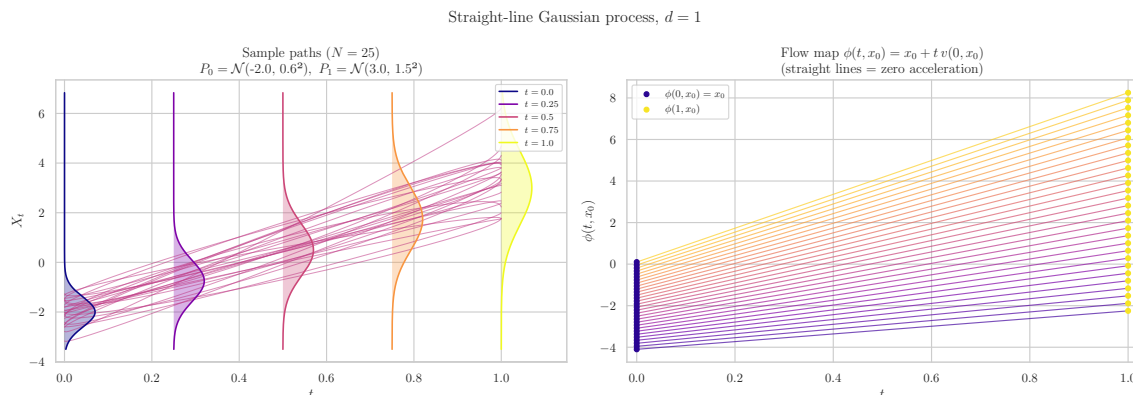


Figure 1: Straight-line Gaussian process for $d = 1$ with endpoints $P_0 = \mathcal{N}(-2, 0.36)$ and $P_1 = \mathcal{N}(3, 2.25)$, as constructed in Proposition [8](#). *Left:* $N = 25$ sample paths X_t in space-time, with marginal Gaussian densities overlaid at $t \in \{0, 0.25, 0.5, 0.75, 1\}$. *Right:* flow map $\phi(t, x_0) = x_0 + t v(0, x_0)$ launched from a regular grid of initial conditions; the trajectories are straight lines, confirming that the pointwise acceleration vanishes identically.

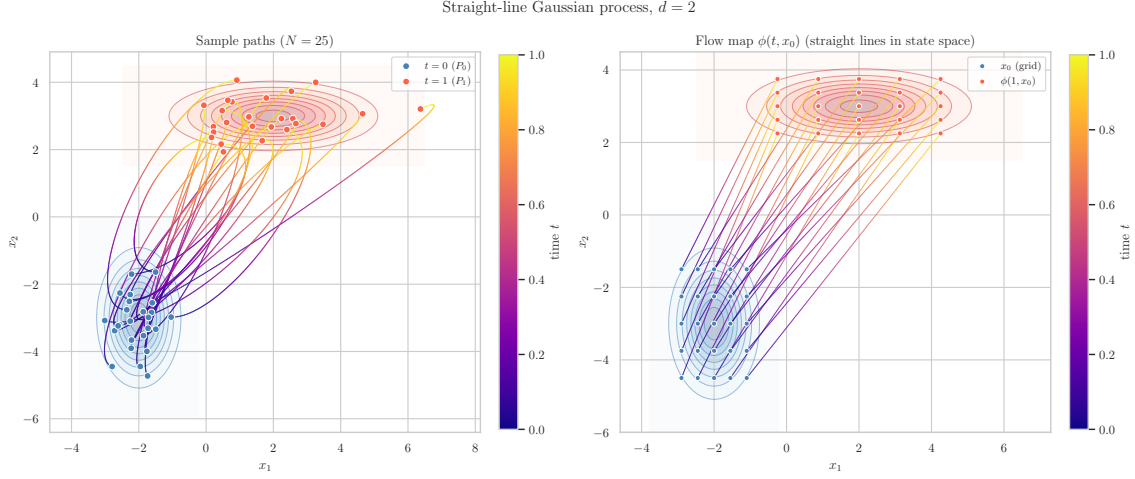


Figure 2: Straight-line Gaussian process for $d = 2$ with diagonal, commuting endpoint covariances $\Sigma_0 = \text{diag}(0.36, 1.0)$ and $\Sigma_1 = \text{diag}(2.25, 0.25)$, as constructed in Theorem 9. Shaded regions show the marginal Gaussian densities of P_0 (blue) and P_1 (red). *Left:* $N = 25$ sample paths in the (x_1, x_2) state space, coloured by time t . *Right:* flow map $\phi(t, x_0)$ launched from an 5×5 grid of initial conditions; the trajectories are straight lines in state space, consistent with zero acceleration.

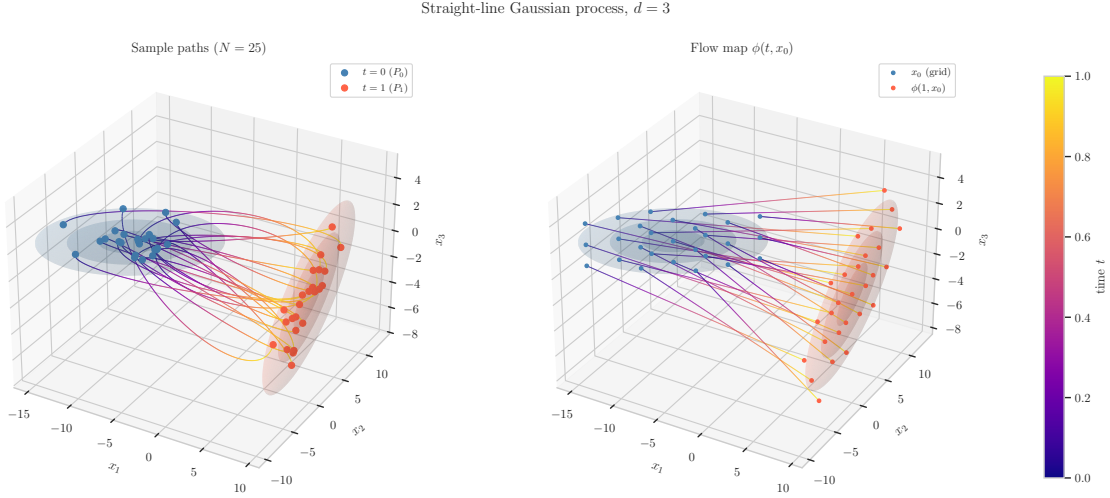


Figure 3: Straight-line Gaussian process for $d = 3$ with non-commutative endpoint covariances. Transparent ellipsoids at the 1σ , 2σ , and 3σ levels mark the supports of P_0 (blue) and P_1 (red). *Left:* $N = 25$ sample paths in (x_1, x_2, x_3) state space, coloured by time t . *Right:* flow map $\phi(t, x_0)$ from a $4 \times 4 \times 4$ grid of initial conditions; straight trajectories confirm zero acceleration throughout.

2 PDE characterization of straight-line flows.

We are concerned with the following problem:

Problem: Given $P_0, P_1 \in \mathcal{P}(\mathbb{R}^d)$, characterize the sets $\mathcal{S}_{\text{SL}}(P_0, P_1)$ and $\mathcal{F}_{\text{SL}}(P_0, P_1)$.

In this section, we make additional regularity assumptions on the flow maps $(\phi_t)_{t \in [0,1]}$ and the processes $X_\bullet$ under consideration. These assumptions allow us to develop an analytical theory of straight-line flows. Recalling the notation from Section 1.3.3 we characterize processes in $\mathcal{S}_{\text{SL}}(P_0, P_1)$ and $\mathcal{F}_{\text{SL}}(P_0, P_1)$ through PDEs involving the conditional statistics v_t , a_t , Π_t , $\mathcal{C}_t$ and the marginal density ρ_t .

We subsequently distinguish a particularly useful characterization of straight-line flows, namely the satisfiability of the PDE

$$D_t v_t = 0,$$

where we recall that D_t is the material derivative associated with the conditional velocity v_t . We use this notion of vanishing material derivative in Section 3 to construct straight-line processes for Gaussian endpoint measures.

Assumption. For the remainder of this section, assume that the marginal densities ρ_t of X_t exist for all $t \in [0, 1]$ and the sample paths $t \mapsto X_t(\omega)$ are in $W^{2,2}([0, 1]; \mathbb{R}^d)$ for each realization $\omega \in \Omega$. Moreover, assume that the trajectory maps $t \mapsto \phi_t(x)$ are in $C^2([0, 1]; \mathbb{R}^d)$ for each $x \in \mathbb{R}^d$, whereas for each $t \in [0, 1]$ the flow map $x \mapsto \phi_t(x)$ are homeomorphisms. Lastly, assume the conditional velocity $v : (t, x) \mapsto v_t(x)$ is a continuously differentiable map jointly in $[0, 1] \times \mathbb{R}^d$.

First, an elementary reformulation allows us to move from the flow ϕ_t to the ensemble velocity.

Proposition 1. Consider the conditional velocity v_t as in Section 1.3.3. For all $(t, x) \in [0, 1] \times \mathbb{R}^d$, one has

$$\partial_{tt} \phi_t(x) = 0 \quad \Longleftrightarrow \quad D_t v_t(x) = 0.$$

See Appendix A for the proof. This result essentially constitutes a passage from the Lagrangian to the Eulerian perspective. Rather than tracking the motion of a single particle via the flow map, the Eulerian perspective considers the total change of the ensemble velocity field.

The second step is to relate the material derivative to other statistical quantities of interest, in particular the second order tensors $\mathcal{C}_t$ and Π_t . Here, a *momentum balance* identity allows us to make progress.

Lemma 2. Consider the conditional velocity v_t , second moment $\mathcal{C}_t$, and acceleration a_t of a process X_t with marginal density ρ_t , as in Section 1.3.3. Then, the following equation holds for all $(t, x) \in [0, 1] \times \mathbb{R}^d$:

$$\partial_t (\rho_t v_t) + \nabla \cdot (\rho_t \mathcal{C}_t) = \rho_t a_t. \quad (6)$$

For the proof, refer to Appendix A. This is a fundamental identity with an elegant physical interpretation. From left to right, one can read the first term as the rate of change of momentum in some elementary volume, the second term is at the net momentum flux, and the third as the net body force acting on said volume.

Finally, inserting the definition of the Reynolds tensor Π_t together with the continuity equation into the above identity, we obtain yet another identity, elucidating the connection between the material derivative and the Reynolds (i.e., covariance) tensor.

Lemma 3. *The following equation holds for all $(x, t) \in [0, 1] \times \mathbb{R}^d$:*

$$\rho_t D_t v_t + \nabla \cdot (\rho_t \Pi_t) = \rho_t a_t .$$

This immediately leads to a corollary and a reformulation of our problem.

Corollary 4. *For all $(t, x) \in [0, 1] \times \mathbb{R}^d$ we have:*

$$D_t v_t = 0 \iff \nabla \cdot (\rho_t \Pi_t) = \rho_t a_t .$$

PDE reformulation: Given $P_0, P_1 \in \mathcal{P}(\mathbb{R}^d)$, characterize all stochastic processes $(X_t)_{t \in [0, 1]}$, with $X_i \sim P_i$ for $i \in \{0, 1\}$, such that

$$\nabla \cdot (\rho_t \Pi_t) = \rho_t a_t , \quad \forall t \in [0, 1] . \tag{7}$$

2.1 Linear characteristics

In this section we demonstrate the usefulness of reformulation (7) to obtain a complete characterization of affine processes that give rise to straight flows. We define an *affine process* with marginals $P_0, P_1 \in \mathcal{P}(\mathbb{R}^d)$ to be a process of the form

$$X_t = (1 - t) X_0 + t X_1 ,$$

where $(X_0, X_1) \sim \gamma$ and $\gamma \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ is a coupling of the measures P_0 and P_1 , meaning that the first and second marginals of γ are P_0 and P_1 , respectively. Although seemingly restrictive, this process is widespread in computational statistics and generative modelling, e.g., optimal-transport displacement interpolation and modern flow-based generative modelling where linear sample-to-sample paths or linear noise-data blends define the training trajectory; see [McC97, LCBH⁺23, Liu22, ABVE23].

The main result of this section is that the flow generated by the ensemble velocity of an affine process can be of straight-line type only if the coupling γ between the endpoints is deterministic, i.e., only if there is some measurable map $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that $T(X_0) = X_1$ almost surely, or equivalently $\gamma = (\text{id} \times T)_\# P_0$, where the operation $\text{id} \times T : \mathbb{R}^d \rightarrow \mathbb{R}^d \times \mathbb{R}^d$ acts as the mapping $x \mapsto (x, T(x))$. In fact, under some mild regularity assumptions on T , this condition is also sufficient.

The key input, here, is that the conditional acceleration $a_t(x) = \mathbb{E}[\ddot{X}_t \mid X_t = x] \equiv 0$ vanishes for affine processes since they have no acceleration at the sample path level, i.e., $\ddot{X}_t \equiv 0$. Thus, Equation (7) yields the straight-line characterization

$$\nabla \cdot (\rho_t \Pi_t) = 0 .$$

Theorem 5. *Suppose that $X_t = (1 - t)X_0 + tX_1$ with $(X_0, X_1) \sim \gamma$ and $\gamma \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ with first and second marginals P_0 and P_1 , respectively. Moreover, suppose that for all $t \in [0, 1]$ we have $\mathbb{E} \|X_t\|^2 < \infty$ and $\mathbb{E} \|\Pi_t(X_t)\|^2 < \infty$ and $t \mapsto \Pi_t(x)$ is continuous at $t = 1$ for all $x \in \mathbb{R}^d$. Then, we have*

$$\nabla \cdot (\rho_t \Pi_t) = 0$$

only if there exists a measurable map $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ satisfying $T_\# P_0 = P_1$ such that

$$(\text{id} \times T)_\# P_0 = \gamma ,$$

where $\text{id} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the identity map. Moreover, the condition is sufficient in $[0, 1)$ provided the Jacobian of the map T is positive semi-definite at any $x \in \mathbb{R}^d$. In particular, this condition holds for a continuously differentiable optimal transport map T satisfying the above conditions on X_t and Π_t .

The proof of Theorem 5 in Appendix A. This result characterizes all affine processes inducing straight flows as those with deterministically coupled endpoints.

3 Existence

Our goal in this section is to describe straight line generalized interpolants whenever the endpoint measures are Gaussian. We begin with the case of when the two multivariate Gaussians have the same covariance matrix, then we build more complicated examples.

3.1 Gaussian endpoint measures

We first describe a straight line generalized interpolant between two Gaussian measures with identical covariances.

Theorem 6. Let $m_0, m_1 \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}_{++}^{d \times d}$ and consider Gaussian measures $P_0 = \mathcal{N}(m_0, \Sigma)$ and $P_1 = \mathcal{N}(m_1, \Sigma)$. Now set

$$F(t, x, y, z) = (1-t)x + ty + \sqrt{2t(1-t)}z \quad \text{and} \quad Q = \mathcal{N}(0, \Sigma).$$

The generalized interpolant (F, Q) satisfies

$$(F, Q) \in \mathcal{F}_{\text{SL}}(P_0, P_1).$$

Proof. The generalized interpolant (F, Q) induces the process

$$X_t = (1-t)X_0 + tX_1 + \sqrt{2t(1-t)}Z.$$

Since all random variables involved are Gaussian, we can compute

$$v(t, X_t) = \mathbb{E}[\dot{X}_t | X_t] = \mathbb{E}[\dot{X}_t] + \text{Cov}(\dot{X}_t, X_t) \text{Var}(X_t)^{-1} (X_t - \mathbb{E}[X_t]),$$

where by $\mathbb{E}[Z] = 0$ we have

$$\mathbb{E}[\dot{X}_t] = m_1 - m_0$$

and by the independence of X_0, X_1, Z we have

$$\begin{aligned} \text{Cov}(\dot{X}_t, X_t) &= \text{Cov}\left(-X_0 + X_1 + \frac{1-2t}{2\sqrt{t(1-t)}}Z, (1-t)X_0 + tX_1 + \sqrt{t(1-t)}Z\right) \\ &= -(1-t)\Sigma + t\Sigma + \frac{1-2t}{\sqrt{2t(1-t)}}\sqrt{2t(1-t)}\Sigma \\ &= (2t-1+1-2t)\Sigma \\ &= 0. \end{aligned}$$

Thus, we conclude that $v(t, x) = m_1 - m_0$ for each $(t, x) \in [0, 1] \times \mathbb{R}^d$ leading to the flow map $\phi(t, x) = x + t(m_1 - m_0)$. In particular, we have that $\partial_t^2 \phi \equiv 0$ thereby completing the proof. $\square$

Corollary 7. For any $m \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}_{++}^{d \times d}$, consider the measure $P = \mathcal{N}(m, \Sigma)$. Then, we have that $\mathcal{F}_{\text{SL}}(P, P) \neq \emptyset$.

Remark 2. At this stage, one might be tempted to generalize the above strategy to pairs of measures

$$P_0 = \mathcal{N}(m_0, \Sigma_0), \quad P_1 = \mathcal{N}(m_1, \Sigma_1),$$

with $m_i \in \mathbb{R}^d$ and distinct $\Sigma_i \in \mathbb{R}_{++}^{d \times d}$ for $i = 0, 1$. Unfortunately, this is not immediately possible. Indeed, consider matrix valued schedules $(A_t)_t, (B_t)_t, (C_t)_t \in \mathbb{R}^{d \times d}$ satisfying the boundary conditions

$$\begin{aligned} A_0 &= I_d, \quad A_1 = 0, \\ B_0 &= 0, \quad B_1 = I_d, \\ C_0 &= 0, \quad C_1 = 0. \end{aligned}$$

and draw $Z \sim \mathcal{N}(m_2, \Sigma_2)$ with $m_2 \in \mathbb{R}^d$ and $\Sigma_2 \in \mathbb{R}_{++}^{d \times d}$. Now set

$$X_t = A_t X_0 + B_t X_1 + C_t Z,$$

and, as done above, attempt to find $(A_t)_t, (B_t)_t, (C_t)_t$ such that $\text{Cov}(\dot{X}_t, X_t) \equiv 0$. Define $\Sigma_t := \text{Var}(X_t) = \text{Cov}(X_t, X_t)$ and by Lemma 24 note that we have $\dot{\Sigma}_t = \text{Cov}(\dot{X}_t, X_t) + \text{Cov}(X_t, \dot{X}_t)$. Since the covariance is symmetric, we obtain

$$\dot{\Sigma}_t \equiv 0.$$

In particular, this implies that $\Sigma_0 = \Sigma_1$ contradicting that the Σ_i are distinct.

Remark 3. Here, we note that deducing that

$$\mathcal{F}_{\text{SL}}(P, P) \neq \emptyset$$

for some $P \in \mathcal{P}(\mathbb{R}^d)$ is not a trivial problem. Indeed, one might expect that the constant process

$$X_t \equiv X_0, \quad \forall t \in [0, 1],$$

lies in $\mathcal{F}_{\text{SL}}(P, P)$, since it satisfies $X_0 \sim P_0$ and $X_1 \sim P_1$ as well as the straight-line condition $D_t v_t \equiv 0$ with the trivial velocity field $v(t, x) = \mathbb{E}[\dot{X}_t | X_t = x] \equiv 0$. The joint law (X_0, X_1) , however, is *not* a product measure, and hence

$$X_\bullet \notin \mathcal{F}_{\text{SL}}(P, P).$$

In fact, the only process $X_\bullet \in \mathcal{F}_{\text{SL}}(P, P)$, for $P = \mathcal{N}(m, \Sigma)$ with $m \in \mathbb{R}^d, \Sigma \in \mathbb{R}_{++}^{d \times d}$, that we are aware of uses the construction of Theorem 6 see Corollary 7

The next step is to study $\mathcal{F}_{\text{SL}}(P_0, P_1)$ when P_0 and P_1 are Gaussian measures with distinct covariance matrices. Here, we may no longer rely on trivial satisfiability of equation (3) by a vanishing covariance; we instead need to use the full equivalence

$$X_\bullet \in \mathcal{F}_{\text{SL}}(P_0, P_1) \iff D_t v_t = \partial_t v_t + (v_t \cdot \nabla) v_t \equiv 0, \quad (8)$$

for a regular enough process $X_\bullet$ with associated velocity field v . To give insight into the proof, we first construct a straight line generalized interpolant for arbitrary univariate Gaussian endpoints and then proceed with the multivariate case.

Proposition 8. Let $P_0 = \mathcal{N}(m_0, \sigma_0^2)$ and $P_1 = \mathcal{N}(m_1, \sigma_1^2)$ be two univariate Gaussian measures with means $m_i \in \mathbb{R}$ and variances $\sigma_i^2 > 0$ for $i = 0, 1$. Set

$$F(t, x, y, z) = (1-t)x + ty + \sqrt{2t(1-t)}z, \\ Q = \mathcal{N}(0, \sigma_0\sigma_1).$$

Then, the generalized interpolant (F, Q) satisfies

$$(F, Q) \in \mathcal{F}_{\text{SL}}(P_0, P_1).$$

Proof. The generalized interpolant (F, Q) induces the process

$$X_t = (1-t)X_0 + tX_1 + \sqrt{2t(1-t)}Z, \quad (9)$$

where $X_0 \sim \mathcal{N}(m_0, \sigma_0^2)$, $X_1 \sim \mathcal{N}(m_1, \sigma_1^2)$, and $Z \sim \mathcal{N}(0, \sigma_0\sigma_1)$ where all random variables are independent. We will verify that the velocity field v associated to $X_\bullet$ satisfies Burger's equation [\[8\]](#). First, using Gaussianity we compute the mean and variance of X_t as

$$\begin{aligned} m_t &:= \mathbb{E}[X_t] \\ &= m_0 + t(m_1 - m_0), \\ \sigma_t^2 &:= \text{Var}(X_t) \\ &= \sigma_0^2(1-t)^2 + \sigma_1^2t^2 + 2t(1-t)\sigma_0\sigma_1 \\ &= (\sigma_0(1-t) + \sigma_1t)^2, \end{aligned}$$

as well as the covariance term

$$\begin{aligned} \text{Cov}(\dot{X}_t, X_t) &= \text{Cov}(X_1 - X_0 + \frac{1-2t}{\sqrt{2t(1-t)}}Z, (1-t)X_0 + tX_1 + \sqrt{2t(1-t)}Z) \\ &= -(1-t)\sigma_0^2 + t\sigma_1^2 + (1-2t)\sigma_0\sigma_1. \end{aligned}$$

Now we notice a key identity:

$$\begin{aligned} \partial_t(\sigma_t^2) &= \partial_t((\sigma_0(1-t) + \sigma_1t)^2) = 2(\sigma_0(1-t) + \sigma_1t)(\sigma_1 - \sigma_0) \\ &= 2(-\sigma_0^2(1-t) + \sigma_1^2t + \sigma_0\sigma_1(1-t) - \sigma_0\sigma_1t) \\ &= 2(-\sigma_0^2(1-t) + \sigma_1^2t + (1-2t)\sigma_0\sigma_1) \\ &= 2\text{Cov}(\dot{X}_t, X_t), \end{aligned}$$

that yields a succinct expression for the velocity field v as

$$\begin{aligned} v(t, x) &:= \mathbb{E}[\dot{X}_t | X_t = x] \\ &= \mathbb{E}[\dot{X}_t] + \text{Cov}(\dot{X}_t, X_t)\text{Var}(X_t)^{-1}(x - \mathbb{E}[X_t]) \\ &= \dot{m}_t + \frac{1}{2} \frac{\partial_t(\sigma_t^2)}{\sigma_t^2}(x - m_t) \\ &= \dot{m}_t + \partial_t \ln \sigma_t(x - m_t) \end{aligned}$$

Plugging into Burger's equation, we have

$$\begin{aligned}
\partial_t v(x, t) + (v \cdot \nabla) v(x, t) &= \partial_t v(x, t) + v(x, t) \partial_x v(x, t) \\
&= \ddot{m}_t + \partial_t^2 \ln \sigma_t (x - m_t) + \partial_t \ln \sigma_t (-\dot{m}_t) + (\dot{m}_t + \partial_t \ln \sigma_t (x - m_t)) (\partial_t \ln \sigma_t) \\
&= \ddot{m}_t + \partial_t^2 \ln \sigma_t (x - m_t) + (\partial_t \ln \sigma_t)^2 (x - m_t) \\
&= \ddot{m}_t + (\partial_t^2 \ln \sigma_t + (\partial_t \ln \sigma_t)^2) (x - m_t).
\end{aligned}$$

We can simplify the factor involving σ_t in the above display as

$$\begin{aligned}
\partial_t^2 \ln \sigma_t + (\partial_t \ln \sigma_t)^2 &= \partial_t \left(\frac{\dot{\sigma}_t}{\sigma_t} \right) + \left(\frac{\dot{\sigma}_t}{\sigma_t} \right)^2 \\
&= \frac{\ddot{\sigma}_t \sigma_t - (\dot{\sigma}_t)^2}{\sigma_t^2} + \left(\frac{\dot{\sigma}_t}{\sigma_t} \right)^2 \\
&= \frac{\ddot{\sigma}_t}{\sigma_t}.
\end{aligned}$$

Note here that since $\sigma_i > 0$ for $i \in \{0, 1\}$ we have $\sigma_t > 0$ for each $t \in [0, 1]$ and thus the above expression is well-defined. Now notice that

$$\ddot{m}(t) \equiv \ddot{\sigma}_t \equiv 0.$$

Indeed, recalling our computations at the beginning of the proof, we have for $t \in [0, 1]$ that

$$\begin{aligned}
\ddot{m}_t &= \partial_t^2 (m_0 + t(m_1 - m_0)) = 0 \\
\ddot{\sigma}_t &= \partial_t^2 (\sigma_0(1 - t) + \sigma_1 t) = 0.
\end{aligned}$$

Therefore, we substitute into the expression for Burger's equation to conclude that

$$\begin{aligned}
\partial_t v_t + (v \cdot \nabla) v &= \ddot{m}_t + (\partial_t^2 \ln \sigma_t + (\partial_t \ln \sigma_t)^2) (x - m_t) \\
&= \ddot{m}_t + \frac{\ddot{\sigma}_t}{\sigma_t} (x - m_t) \\
&= (0) + (0) (x - m_t) \\
&= 0.
\end{aligned}$$

By Proposition [1](#) we conclude that the given path X_t satisfies the straight-line flow. $\square$

Remark 4. The essence of the above argument above is the equivalence

$$D_t v_t \equiv 0 \iff \begin{cases} \ddot{m}_t \equiv 0, \\ \ddot{\sigma}_t \equiv 0. \end{cases}$$

in the case of Gaussian marginals $X_t \sim \mathcal{N}(m_t, \sigma_t^2)$. The construction of F and Q is precisely such that the system on the right is satisfied. Lastly, we note that having the triple (X_0, X_1, Z) be jointly Gaussian was essential since it allowed us to use the explicit formulas for Gaussian conditioning.

Theorem 9. Let $P_0 = \mathcal{N}(m_0, \Sigma_0)$ and $P_1 = \mathcal{N}(m_1, \Sigma_1)$ be two nondegenerate Gaussian measures with means $m_i \in \mathbb{R}^d$ and $\Sigma_i \in \mathbb{R}_{++}^{d \times d}$ positive definite matrices for $i = 0, 1$. Then, there is diagonal Λ and matrix V such that

$$\Sigma_0 = VV^\top, \quad \Sigma_1 = V\Lambda V^\top.$$

For $\Sigma_Z = V\Lambda^{1/2}V^\top$, we know that $\Sigma_Z \in \mathbb{R}_{++}^{d \times d}$. Further, the generalized interpolant defined via

$$F(t, x, y, z) = (1-t)x + ty + \sqrt{2t(1-t)}z, \\ Q = \mathcal{N}(0, \Sigma_Z).$$

satisfies

$$(F, Q) \in \mathcal{F}_{\text{SL}}(P_0, P_1).$$

Proof. We remark that, because both covariance matrices are positive definite, the pencil (Σ_0, Σ_1) is also positive definite—the pencil’s eigenvalues are shared by the matrix $\Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}$ (using the symmetric square root $\Sigma_0^{1/2}$), which is also SPD. Therefore, we can take Λ as the diagonal matrix of (positive) eigenvalues of the pencil. Similarly, the pencil has nonsingular eigenvector matrix $\tilde{V}$ defined such that $\Sigma_1 = \Sigma_0\tilde{V}\Lambda\tilde{V}^{-1}$ and $\tilde{V}^\top\Sigma_0\tilde{V} = I$. Thus, we know $V = \tilde{V}^{-\top}$ satisfies the identity $\Sigma_0 = VV^\top$. We observe that Σ_1 has a similar property:

$$V^{-1}\Sigma_1 = \tilde{V}^\top\Sigma_1 = \tilde{V}^\top\Sigma_0\tilde{V}\Lambda\tilde{V}^{-1} = \Lambda\tilde{V}^{-1} = \Lambda V^\top.$$

Our diagonal matrix Λ is thus positive and V satisfies the constraints, making Σ_Z symmetric positive definite. Now we define the interpolant

$$X_t = (1-t)X_0 + tX_1 + \sqrt{2t(1-t)}Z, \quad X_i \sim \mathcal{N}(m_i, \Sigma_i), \quad i = 0, 1, \quad Z \sim \mathcal{N}(0, \Sigma_Z),$$

where the auxiliary covariance matrix Σ_Z is defined as in the theorem statement. Defining matrix $\Xi = \Lambda^{1/2} - I$, we find that

$$\begin{aligned} \Sigma_t &= (1-t)^2\Sigma_0 + t^2\Sigma_1 + 2t(1-t)\Sigma_Z \\ &= V \left((1-t)^2I + t^2\Lambda + 2t(1-t)\Lambda^{1/2} \right) V^\top \\ &= V((1-t)I + t\Lambda^{1/2})^2 V^\top \\ &= V(I + t\Xi)^2 V^\top \\ \dot{\Sigma}_t &= 2V(I + t\Xi)\Xi V^\top \\ \ddot{\Sigma}_t &= 2V\Xi^2 V^\top. \end{aligned}$$

Further, one can find that

$$\begin{aligned} \text{Cov}(\dot{X}_t, X_t) &= \text{Cov} \left(X_1 - X_0 + \frac{1-2t}{\sqrt{2t(1-t)}}Z, (1-t)X_0 + tX_1 + \sqrt{2t(1-t)}Z \right) \\ &= -(1-t)\Sigma_0 + t\Sigma_1 + (1-2t)\Sigma_Z \\ &= V(-(1-t)I + t\Lambda + (1-2t)\Lambda^{1/2})V^\top \\ &= V(\Xi + t\Xi^2)V^\top \\ &= V(I + t\Xi)\Xi V^\top \\ &= \frac{1}{2}\dot{\Sigma}_t. \end{aligned}$$

Then, we express the conditional velocity field as

$$\begin{aligned} v(t, x) &= \mathbb{E}[\dot{X}_t | X_t = x] \\ &= \mathbb{E}[\dot{X}_t] + \text{Cov}(\dot{X}_t, X_t) \text{Var}(X_t)^{-1} (x - \mathbb{E}[X_t]) \\ &= \dot{m}_t + \frac{1}{2} \dot{\Sigma}_t \Sigma_t^{-1} (x - m_t). \end{aligned}$$

With the intention of making Burger's equation (8) explicit in this example, we compute

$$\begin{aligned} 2\partial_t v &= 2\ddot{m}_t + \ddot{\Sigma}_t \Sigma_t^{-1} (x - m_t) - \dot{\Sigma}_t \Sigma_t^{-1} \dot{\Sigma}_t \Sigma_t^{-1} (x - m_t) + \dot{\Sigma}_t \Sigma_t^{-1} (-\dot{m}_t) \\ &= 2\ddot{m}_t + (\ddot{\Sigma}_t + \dot{\Sigma}_t \Sigma_t^{-1} \dot{\Sigma}_t) \Sigma_t^{-1} (x - m_t) - \dot{\Sigma}_t \Sigma_t^{-1} \dot{m}_t. \end{aligned}$$

Similarly, we calculate $\nabla v = \frac{1}{2} \dot{\Sigma}_t \Sigma_t^{-1}$, giving

$$2(v \cdot \nabla)v = \dot{\Sigma}_t \Sigma_t^{-1} \dot{m}_t + \frac{1}{2} \dot{\Sigma}_t \Sigma_t^{-1} \dot{\Sigma}_t \Sigma_t^{-1} (x - m_t).$$

Using these above computations, we can rewrite Burger's equation (8) as

$$2(\partial_t v + (v \cdot \nabla)v) = 2\ddot{m}_t + \left(\ddot{\Sigma}_t - \frac{1}{2} \dot{\Sigma}_t \Sigma_t^{-1} \dot{\Sigma}_t \right) \Sigma_t^{-1} (x - m_t).$$

We know, however, that mean m_t has the form $m_t = (1-t)m_0 + tm_1$, making $\ddot{m}_t \equiv 0$ by construction. Further, we can expand the covariance term to see that

$$\begin{aligned} \frac{1}{2} \dot{\Sigma}_t \Sigma_t^{-1} \dot{\Sigma}_t &= 2V(I + t\Xi)\Xi(I + t\Xi)^{-2}(I + t\Xi)\Xi V^\top \\ &= 2V\Xi^2 V^\top \\ &\equiv \ddot{\Sigma}_t, \end{aligned}$$

where the cancellations are due to the commutativity of diagonal matrices. Therefore, we can conclude $\ddot{\Sigma}_t - \frac{1}{2} \dot{\Sigma}_t \Sigma_t^{-1} \dot{\Sigma}_t \equiv 0$ and, by Proposition 1 the process $X_\bullet$ must satisfy the straight-line flow condition. $\square$

Remark 5. Note that, in the case of commuting covariance matrices Σ_0, Σ_1 , we can define matrices $S_i = \Sigma_i^{1/2}$ as the symmetric square roots of the respective covariance matrices for $i = 0, 1$. Then, one can show $\Sigma_Z = S_0 S_1$, a clearer extension of the geometric mean from the one-dimensional case.

Remark 6. For a Brownian motion $B_\bullet$, the standard Brownian bridge is defined by $W_t = B_t - t B_1$. Then, for any $P_0, P_1 \in \mathcal{P}(\mathbb{R}^d)$, we draw $(X_0, X_1) \sim P_0 \otimes P_1$ to define the Brownian bridge with independent endpoints by

$$W_t^{P_0, P_1} = (1-t)X_0 + tX_1 + W_t.$$

Notice the similarity of the above display to the construction in Theorem 9. The key difference is that the correction term W_t has covariance $t(1-t)I_d$ instead of $2t(1-t)\Sigma_Z$, with I_d the d -dimensional identity matrix.

3.2 Trivial solutions with non-Lipschitz velocity

In this section we discuss the existence of a seemingly trivial almost everywhere solution to the PDE (7) that does not, however, satisfy one of the key assumptions of Definition 4. We discuss this here for two reasons: First, it demonstrates the necessity of the Lipschitz assumption made in Definition 4. Second, it serves as a warning for the reader wishing to extend our work.

Proposition 10. *Choose any $\tau \in (0, 1)$ and consider the process X_t such that*

$$X_t = \alpha_t X_0 + \beta_t X_1, \quad \alpha_t = \begin{cases} 1 - \frac{t}{\tau} & t < \tau \\ 0 & t \geq \tau \end{cases}, \quad \beta_t = \begin{cases} 0 & t \leq \tau \\ \frac{t-\tau}{1-\tau} & t > \tau \end{cases}.$$

Then, sample trajectories X_t satisfy PDE (7) for all $t \in [0, 1] \setminus \{\tau\}$, but the associated velocity field v does not satisfy the Lipschitz assumption in Definition 4. Therefore, our paths are not straight, i.e., $X_\bullet \notin \mathcal{S}_{\text{SL}}(P_0, P_1)$ for any P_0, P_1 .

Proof. For the remainder of the proof, suppose $t \neq \tau$. First, consider the case $t < \tau$. We have

$$\begin{aligned} v_t(x) &= \mathbb{E}[\dot{X}_t | X_t = x] \\ &= \mathbb{E}\left[-\frac{1}{\tau} X_0 \mid \left(1 - \frac{t}{\tau}\right) X_0 = x\right] \\ &= -\frac{1}{\tau} \frac{\tau}{\tau - t} \mathbb{E}\left[\frac{\tau - t}{\tau} X_0 \mid \frac{\tau - t}{\tau} X_0 = x\right] \\ &= \frac{x}{t - \tau}. \end{aligned}$$

using linearity in the third line and the definition of the conditional expectation in the fourth. Using similar reasoning, we can compute

$$\begin{aligned} \mathcal{C}_t(x) &:= \mathbb{E}[\dot{X}_t \otimes \dot{X}_t | X_t = x] \\ &= \frac{1}{\tau^2} \mathbb{E}\left[X_0 \otimes X_0 \mid \left(1 - \frac{t}{\tau}\right) X_0 = x\right] \\ &= \frac{x \otimes x}{(t - \tau)^2} \\ &= v_t(x) \otimes v_t(x). \end{aligned}$$

Similarly, for the case $t > \tau$ we compute

$$\begin{aligned} v_t(x) &= \mathbb{E}[\dot{X}_t | X_t = x] \\ &= \mathbb{E}\left[\frac{1}{1 - \tau} X_1 \mid \frac{t - \tau}{1 - \tau} X_1 = x\right] \\ &= \frac{1}{1 - \tau} \frac{1 - \tau}{t - \tau} \mathbb{E}\left[\frac{t - \tau}{1 - \tau} X_1 \mid \frac{t - \tau}{1 - \tau} X_1 = x\right] \\ &= \frac{x}{t - \tau}. \end{aligned}$$

and also

$$\begin{aligned}
\mathcal{C}_t(x) &:= \mathbb{E}[\dot{X}_t \otimes \dot{X}_t | X_t = x] \\
&= \frac{1}{(1-\tau)^2} \mathbb{E} \left[X_1 \otimes X_1 \mid \frac{t-\tau}{1-\tau} X_1 = x \right] \\
&= \frac{x \otimes x}{(t-\tau)^2} \\
&= v_t(x) \otimes v_t(x).
\end{aligned}$$

Thus, for all $t \in [0, 1] \setminus \{\tau\}$ we have

$$\Pi_t(x) = \mathcal{C}_t(x) - v_t(x) \otimes v_t(x) \equiv 0.$$

On the other hand, the acceleration field $a_t(x)$ is trivially vanishing

$$\begin{aligned}
a_t(x) &= \mathbb{E}[\ddot{X}_t | X_t = x] \\
&= \mathbb{E}[0 | X_t = x] \equiv 0.
\end{aligned}$$

Therefore, the PDE (7) is trivially satisfied for all $t \in [0, 1] \setminus \{\tau\}$. The velocity field $v_t(x) = \frac{x}{t-\tau}$, however, is not Lipschitz in x uniformly in t . Indeed, for any $x \neq y \in \mathbb{R}^d$ note that

$$\frac{\|v_t(x) - v_t(y)\|}{\|x - y\|} = \frac{1}{|t - \tau|},$$

and the right hand side diverges as $t \rightarrow \tau$. □

Remark 7. Note that the above process can easily be realized by a generalized interpolant with $F(t, x, y, z) = \alpha_t x + \beta_t y$ and $Q = \delta_0$ the Dirac mass at zero. Therefore, if this process *did* satisfy the requisite Lipschitz assumption, it would be a straight-line interpolant between any two measures P_0 and P_1 .

Remark 8. Intuitively, the above process $X_\bullet$ collapses all points to the origin at time $t = \tau$ and then re-expands them. For both $t < \tau$ and $t > \tau$ no two characteristics cross and therefore the Reynolds tensor Π_t vanishes. Indeed, this flow is sometimes referred to as *monokinetic*.

4 Impossibility

Above, we constructed straight-line generalized interpolants for multi-variate Gaussian endpoint measures. Next, we investigate the existence of generalized interpolants and, more generally, straight-line processes for arbitrary end-point measures $P_0, P_1 \in \mathcal{P}(\mathbb{R}^d)$. We show that

$$\mathcal{F}_{\text{SL}} \subseteq \mathcal{S}_{\text{SL}} = \emptyset,$$

see (4) and (5) for definitions, when measures P_0, P_1 are “multi-modal” in a sense that will be made precise. To arrive at our main result we present three theorems that investigate this problem at increasing levels of generality. Specifically, we start with the case of $P_0 = P_1 = P$ and P has disconnected support in $d = 1$, see Theorem 11. Then, we generalize to $d \in \mathbb{N}$, see Theorem 13. We finish with the case of $P_0 \neq P_1$ supported on the entire real line $\mathbb{R}$, with P_0, P_1 concentrated appropriately, see Definition 6.

4.1 Disconnected support

As a first step, we show that for any $P \in \mathcal{P}_{\text{a.c.}}(\mathbb{R}^d)$ with disconnected support the set $\mathcal{S}_{\text{SL}}(P, P)$ defined in (4) is empty. This serves as a good way to introduce the proof techniques used in the main theorem of the section. Moreover, it builds on Remark 3 in the previous section, where we noted that constructing straight line processes between identical endpoint measures is nontrivial. In fact, the theorem below shows it can be impossible.

Definition 5. A measure $P \in \mathcal{P}(\mathbb{R}^d)$ is has *disconnected support* if there exist bounded open sets $S_0, S_1 \subseteq \mathbb{R}^d$ such that $\overline{\text{conv}} S_0 \cap \overline{\text{conv}} S_1 = \emptyset$, $\text{supp}(P) = \overline{S_0} \cup \overline{S_1}$ with $P(S_i) > 0$ for $i = 0, 1$ —see Section 1.3.2 for the meaning of $\overline{\text{conv}}$. Furthermore, we say P has a *properly* disconnected support if, additionally, for each $i \in \{0, 1\}$ and any Borel set $A \subseteq S_i$ with positive Lebesgue measure we have $P(A) > 0$.

Remark 9. We note that defining a *disconnected support* as $\overline{\text{conv}} S_0 \cap \overline{\text{conv}} S_1 = \emptyset$ as opposed to $S_0 \cap S_1 = \emptyset$ is justified for this work. We describe a measure where the support is “well-separated” in the sense that it lives on disjoint sets that are “far away” from each other. Examples of measures with connected support $\overline{\text{conv}} S_0 \cap \overline{\text{conv}} S_1 \neq \emptyset$ but empty intersection $S_0 \cap S_1 = \emptyset$ include intertwined and thickened semi-circles, which we wish to rule out. While the assumption of $\overline{\text{conv}} S_0 \cap \overline{\text{conv}} S_1 = \emptyset$ is necessary for our proof of Theorem 13, it can be circumvented in the proof of Theorem 11 by introducing more technicalities and emulating the technique used in the proof of Theorem 17.

Theorem 11. *Suppose that $P \in \mathcal{P}(\mathbb{R})$ has properly disconnected support. Then, we have*

$$\mathcal{S}_{\text{SL}}(P, P) = \emptyset.$$

Remark 10. The proof is illustrated in Figure 4.

Proof of Theorem 11. Let S_0, S_1 be as in Definition 5 with $d = 1$. By shifting the origin if necessary, we can assume without loss of generality that

$$\sup S_0 =: s_0 < 0 < s_1 =: \inf S_1. \quad (10)$$

Suppose $\mathcal{S}_{\text{SL}}(P, P) \neq \emptyset$ and take any $X_\bullet \in \mathcal{S}(P, P)$. We will work towards a contradiction.

Step 1: Flow is continuous & globally injective

Suppose that $v : [0, 1] \times \mathbb{R} \rightarrow \mathbb{R}$ is the velocity field associated to the process $X_\bullet$, see Definition 4 and ϕ is the flow map driven by v . Now since

$$\partial_t^2 \phi(t, x) = 0,$$

by Proposition 19 we have that

$$\phi(t, x) = x + t v(0, x).$$

Because v satisfies the assumptions of Definition 4 then by [Tes12, Theorem 2.5] we have

$$x \neq x' \implies \phi(t, x) \neq \phi(t, x') \quad \text{for all } t \in [0, 1]. \quad (11)$$

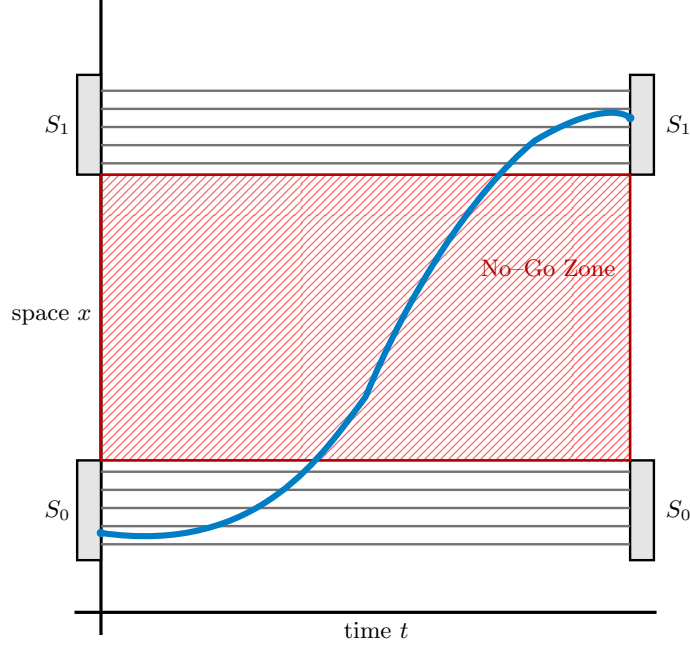


Figure 4: Schematic depiction of the proof of Theorem 11

Indeed, the initial value problem (IVP)

$$\begin{cases} \frac{d}{ds}y(s) = v(s, y(s)) \\ y(t) = z, \end{cases}$$

admits unique local solutions for every $(t, z) \in [0, 1] \times \mathbb{R}$. If $\phi(t, x) = \phi(t, x')$ for some $t_\star \in [0, 1]$ and $x \neq x'$ then the curves

$$y(t) = \phi(t, x), \quad y'(t) = \phi(t, x'),$$

are both local solutions to the above IVP with $t = t_\star$ and $z = \phi(t_\star, x) = \phi(t_\star, x')$ so they coincide in a neighborhood of $t_\star$. By a continuation argument, it must be $y(t) = y'(t)$ for all $t \in [0, 1]$ and so in particular $y(0) = y'(0)$ implying that $x = x'$, contradicting our assumption.

Finally, let us note that, for each $t \in [0, 1]$, the flow map $\phi(t, \cdot) : \mathbb{R} \rightarrow \mathbb{R}$ is continuous. This follows from [Tes12 Theorem 2.9] and the assumed properties of v , see Definition 4. Note that the aforementioned theorem furnishes continuity of the two-point flow map that solves the above IVP from time s to a later time t , which is combined with global existence on $[0, 1]$, local uniqueness and a standard patching argument to obtain continuity of the map $x \mapsto \phi(t, x)$ for any $t \in [0, 1]$.

Step 2: Topological Obstruction

We claim that for P -almost every $x \in S_i$ we have that

$$\phi(t, S_i) \subseteq \overline{\text{conv}} S_i \text{ for } i \in \{0, 1\}. \quad (12)$$

and all $t \in [0, 1]$. In fact, since S_i is open, $\phi(t, \cdot) : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous map for each $t \in [0, 1]$ (see **Step 1**) and $\overline{\text{conv}} S_i$ is closed, it follows that the above display holds if and only if it holds for all $x \in S_i$.² Notice, here, that since the $\overline{\text{conv}} S_i$ are convex sets and by **Step 1** the paths $t \mapsto \phi(t, x)$ are lines, for all x , it follows that

$$\phi(t, S_i) \subseteq \overline{\text{conv}} S_i \iff \phi(1, S_i) \subseteq \overline{\text{conv}} S_i \quad \text{for } i \in \{0, 1\}.$$

Now by Corollary 26 we have that

$$\phi(1, \cdot)_\# P = P,$$

and therefore it must be that

$$\phi(1, S_i) \subseteq \overline{\text{conv}} S_0 \cup \overline{\text{conv}} S_1 \quad \text{for } i \in \{0, 1\}. \quad (13)$$

Indeed, if $x \in S_i$ satisfies $\phi(1, x) \notin \overline{\text{conv}} S_0 \cup \overline{\text{conv}} S_1$ then, by continuity of $x \mapsto \phi(1, x)$, there is a neighborhood $x \in U \subseteq S_i$ such that $\phi(1, U) \cap (\overline{\text{conv}} S_0 \cup \overline{\text{conv}} S_1) = \emptyset$. Using the properness of the support of P , see Definition 5, we conclude that $\phi(1, \cdot)_\# P$ has support on a set disjoint from $\overline{\text{conv}} S_0 \cup \overline{\text{conv}} S_1$ and therefore it cannot be equal to P . Finally, recalling that the set S_i is connected and the map $x \mapsto \phi(1, x)$ is continuous and using that the sets $\overline{\text{conv}} S_0$ and $\overline{\text{conv}} S_1$ are disjoint and by equation 13 the set $\phi(1, S_i)$ lies in their union, it must be that $\phi(1, S_i) \subseteq \overline{\text{conv}} S_j$ for some $j \in \{0, 1\}$, see [Mun00] Lemma 23.2]. By the global injectivity of the flow, it must be that $\phi(1, S_i) \subseteq \overline{\text{conv}} S_i$ for $i \in \{0, 1\}$, proving the expression 12.

Step 3: A no-go zone

Using 12, we have that

$$\phi(t, S_i) \subseteq \overline{\text{conv}} S_i \quad \text{for } i \in \{0, 1\}.$$

Hence, for all $x \in \mathbb{R}^d$, we have

$$\phi(t, x) \notin G := (s_0, s_1). \quad (14)$$

Here, recall that $\sup S_0 = s_0 < 0 < \inf S_1 = s_1$ ³ and we call G the *no-go zone*. Now, by Corollary 26, the process $X_\bullet \in \mathcal{S}(P_0, P_1)$ satisfies the distributional equality

$$\phi(t, X_0) \stackrel{d}{=} X_t.$$

Combining this equivalence of distributions with 14, we have that

$$\mathbb{P}(X_t \notin G) \quad \text{for each } t \in [0, 1], \quad (15)$$

and intersecting over the rationals $\mathbb{Q}$ we get

$$\mathbb{P}(\forall t \in \mathbb{Q} \cap [0, 1], X_t \notin G) = 1.$$

²For any open set $S \subset \mathbb{R}^d$, closed set $A \subseteq \mathbb{R}^d$ and continuous map $f : S \rightarrow \mathbb{R}^d$, if $f(x) \in A$ for Lebesgue a.e. $x \in S$ then $f(x) \in A$ for all $x \in S$. Indeed, the set $E = \{x \in S : f(x) \notin A\}$ is open and has zero Lebesgue measure, so it must be empty.

³Here, we are using the following well-known representation: $x \in \overline{\text{conv}} S$ if and only if x lies in the closure of $\text{conv} S$ and $z \in \text{conv} S$ if there exist points $y_1, \dots, y_k \in S$ and weights $\lambda_1, \dots, \lambda_k \in [0, 1]$ with $\sum_{i=1}^k \lambda_i = 1$ such that $z = \sum_{i=1}^k \lambda_i y_i$. In particular, if S is a subset of $\mathbb{R}$ then $\overline{\text{conv}} S \subseteq [\inf S, \sup S]$.

By continuity of the sample paths and openness of G , if $\exists \omega \in \Omega$ such that $X_t(\omega) \in G$ for some $t \in [0, 1]$ then there exists an open set $V \subset [0, 1]$ such that $X_s(\omega) \in G$ for all $s \in V$. By density of $\mathbb{Q}$ in $\mathbb{R}$ there exists $t' \in V \cap \mathbb{Q}$ such that $X_{t'}(\omega) \in G$. It follows that

$$\mathbb{P}(\exists t \in [0, 1], X_t \in G) \leq \mathbb{P}(\exists t \in \mathbb{Q} \cap [0, 1], X_t \in G) = 0,$$

which is equivalent to

$$\mathbb{P}(\forall t \in [0, 1], X_t \notin G) = 1. \quad (16)$$

Step 4: Contradiction

Finally, note that with probability $P_1(S_0) \cdot P_1(S_1)$ we sample a pair $(X_0, X_1) \in S_0 \times S_1$. Now by continuity of the sample paths of $X_\bullet$ and the intermediate value theorem, there exists $t^* \in (0, 1)$ such that

$$X_{t^*} \in G = (s_0, s_1).$$

In other words, we have shown that

$$\mathbb{P}(\exists t \in [0, 1], X_t \in G) \geq P_1(S_0) \cdot P_1(S_1) > 0,$$

contradicting equation (16) and completing the proof. $\square$

Remark 11. At this stage, one might wonder whether the impossibility is due to selecting mixture models for both endpoints. In Section 4.2 we see that this is not the case.

As a concrete example of the above theorem, we have the following corollary.

Corollary 12. *Consider the mixture of uniform measures*

$$P = \text{Unif}\left(-\frac{3}{2}, -\frac{1}{2}\right) + \text{Unif}\left(\frac{1}{2}, \frac{3}{2}\right).$$

Then, we have that

$$\mathcal{S}_{\text{SL}}(P, P) = \emptyset,$$

that is, there is no straight-line process connecting P to itself.

Remark 12. The proof of Theorem 11 can be extended to arbitrary dimension d , see Theorem 13. When generalizing to measures with connected but appropriately concentrated support, however, we must replace the topological argument above with a more quantitative proof strategy. See the proof of Theorem 17 for details.

Theorem 13. *Suppose $P \in \mathcal{P}(\mathbb{R}^d)$ has disconnected support on the sets S_0 and S_1 , see Definition 5 and further suppose that*

$$P(S_0) \neq P(S_1).$$

Then, we have

$$\mathcal{S}_{\text{SL}}(P, P) = \emptyset$$

Proof. **Step 1** and **Step 4** in the proof of Theorem 11 extend verbatim. Therefore, it suffices to provide an alternate no-go zone $G \subseteq \mathbb{R}^d$ satisfying the following properties:

1. $G \subseteq \mathbb{R}^d$ is an open set,
2. We have $\mathbb{P}(X_t \in G) = 0$ for all $t \in [0, 1]$ and,
3. For any (deterministic) path $\gamma \in C^0([0, 1]; \mathbb{R}^d)$ such that $\gamma(0) \in S_0$ and $\gamma \in S_1$ there is a $0 < t < 1$ such that $\gamma(t) \in G$.

We claim that the set

$$G = \mathbb{R}^d \setminus (\overline{\text{conv}} S_0 \cup \overline{\text{conv}} S_1) ,$$

satisfies the above properties. Indeed, since the sets $\overline{\text{conv}} S_0$ and $\overline{\text{conv}} S_1$ are closed the set G is open. Moreover, assume any path $\gamma \in C^0([0, 1]; \mathbb{R}^d)$ such that $\gamma(0) \in S_0$ and $\gamma(1) \in S_1$ does not enter G at any $t \in [0, 1]$. Define the last exit and first entry times of $\overline{\text{conv}} S_0$ and $\overline{\text{conv}} S_1$ respectively as

$$t_0 = \sup\{t \in [0, 1] : \gamma(t) \in \overline{\text{conv}} S_0\} , \quad t_1 = \inf\{t \in [0, 1] : \gamma(t) \in \overline{\text{conv}} S_1\} ,$$

and note that $\gamma(t) \notin G$ for all t implies that $t_0 = t_1$. But now for any $\epsilon > 0$ we have $\gamma(t_0 - \epsilon) \in \overline{\text{conv}} S_0$ and $\gamma(t_1 + \epsilon) \in \overline{\text{conv}} S_1$ therefore

$$|\gamma(t_0 - \epsilon) - \gamma(t_1 + \epsilon)| \geq \text{dist}(\overline{\text{conv}} S_0, \overline{\text{conv}} S_1) > 0 .$$

In other words,

$$\lim_{t \rightarrow t_0+} \gamma(t) \neq \lim_{t \rightarrow t_0-} \gamma(t)$$

which contradicts the continuity of γ , completing the proof of property 3.

Finally, for property 2 we argue as follows: recall that by **Step 1** we have the straight-line representation of the flow map

$$\phi(t, x) = (1 - t)x + t\phi(1, x) .$$

Now, notice that we must have

$$\phi(1, S_0) \subseteq \overline{\text{conv}} S_0 \cup \overline{\text{conv}} S_1 , \tag{17}$$

which follows from the following facts: $x \mapsto \phi(1, x)$ is continuous, **Step 1** in the proof of Theorem [11](#) $\phi(1, \cdot)_\# P = P$ and P is supported on $S_0 \cup S_1$, see **Step 2** in the proof of Theorem [11](#) for details. Moreover, again by the continuity of $\phi(1, \cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and the connectedness of each set S_i , it follows that the set $\phi(1, S_0)$ must be also connected, see [Mun00](#), Lemma 23.2]. Since $\overline{\text{conv}} S_0$ and $\overline{\text{conv}} S_1$ are disconnected and equation [17](#) holds, it follows that

$$\phi(1, S_0) \subseteq \overline{\text{conv}} S_0 \quad \text{or} \quad \phi(1, S_0) \subseteq \overline{\text{conv}} S_1 .$$

The latter, however, is impossible since $P(S_0) \neq P(S_1)$ and $\phi(1, \cdot)_\# P = P$. Therefore, we have

$$\phi(1, S_0) \subseteq \overline{\text{conv}} S_0 ,$$

which by convexity implies that

$$\phi(t, S_0) \subseteq \overline{\text{conv}} S_0 ,$$

for all $t \in [0, 1]$. Mutatis mutandis, we can show that for all $t \in [0, 1]$ we have

$$\phi(t, S_1) \subseteq \overline{\text{conv}} S_1 . \tag{18}$$

Finally, recalling the distributional equality

$$X_t \stackrel{d}{=} \phi(t, X_0), \quad X_0 \sim P$$

discussed in **Step 3** of the proof of Theorem [11](#) we obtain that

$$\mathbb{P}(X_t \in G) = 0 \text{ for all } t \in [0, 1],$$

as $\phi(t, X_0)$ has no support on $G = \mathbb{R}^d \setminus (\overline{\text{conv}} S_0 \cup \overline{\text{conv}} S_1)$. $\square$

Remark 13. While we intuit that the result in Theorem [13](#) holds for equal probability $P(S_0) = P(S_1)$, this case is currently not covered by the above argument.

Remark 14. We close this section by noting that decomposing the measure in exactly two disconnected components S_0, S_1 satisfying the assumptions of Definition [5](#) is not essential. In fact, one can easily generalize the above arguments to a measure P supported on any $N \in \mathbb{N}$ components $S_1, \dots, S_N$ for both the $d = 1$ and $d > 1$ cases. One imposes the natural assumptions, e.g., each S_i is open and bounded, for each $i \neq j$ we have $\overline{\text{conv}} S_i \cap \overline{\text{conv}} S_j = \emptyset$ etc., and in the $d > 1$ case the additional condition that $P(S_i) \neq P(S_j)$ for $i \neq j$.

4.2 Connected support in $d = 1$

At this stage, one might wonder whether the impossibility above relies on the disjoint supports of the modes of the mixture model. In this section, we demonstrate that this is not the case. Indeed, we can get a similar impossibility result for target measures with ϵ -disconnected supports.

Definition 6. A measure $P \in \mathcal{P}(\mathbb{R}^d)$ has ϵ -disconnected support if there exist bounded open sets $S_0, S_1 \subseteq \mathbb{R}^d$ such that $\overline{\text{conv}} S_0 \cap \overline{\text{conv}} S_1 = \emptyset$, $P(S_i) > 0$ for $i = 0, 1$ and

$$P(S_0) + P(S_1) \geq 1 - \epsilon.$$

Furthermore, we say P has a *properly* ϵ -disconnected support if in addition to the above the following positivity condition holds: for each $i \in \{0, 1\}$ and any Borel set $A \subseteq S_i$ with positive Lebesgue measure, we have $P(A) > 0$.

Remark 15. The arguments in this section only require $\overline{S_0} \cap \overline{S_1} = \emptyset$. To stay consistent with Remark [9](#) however, we opt to continue using $\overline{\text{conv}} S_0 \cap \overline{\text{conv}} S_1 = \emptyset$.

Indeed, similar impossibility results hold for standard normal source measure and a certain bi-modal Gaussian mixture target measure. To that end, we introduce a useful object followed by a technical lemma.

Definition 7. The *modulus of continuity* of a function $f : [0, 1] \rightarrow \mathbb{R}$ is defined as

$$\kappa_f(\delta) = \sup_{\substack{t, s \in [0, 1] \\ |t - s| \leq \delta}} |f(t) - f(s)|.$$

The modulus of continuity of the stochastic process $X_\bullet : [0, 1] \rightarrow \mathbb{R}$ is the random function $\kappa_{X_\bullet} : (0, 1] \rightarrow [0, \infty)$ with realizations

$$\kappa_{X_\bullet}(\delta; \omega) = \sup_{\substack{t, s \in [0, 1] \\ |t - s| \leq \delta}} |X_t(\omega) - X_s(\omega)|,$$

for each $\omega \in \Omega$.

Remark 16. As is customary with random objects, we will often write $\kappa_{X_\bullet}(\delta)$ for the random variable $\omega \mapsto \kappa_{X_\bullet}(\delta; \omega)$ or $\kappa_{X_\bullet}$ for the random function $\omega \mapsto (\delta \mapsto \kappa_{X_\bullet}(\delta; \omega))$.

Definition 8. For positive reals $A, \alpha, \beta > 0$, let $\mathcal{C}(A, \alpha, \beta)$ denote the class of stochastic processes $X_\bullet$ with continuous sample paths satisfying

$$\mathbb{P}(\kappa_{X_\bullet}(\delta) \geq \theta) \leq A \frac{\delta^\alpha}{\theta^\beta},$$

for all $\theta, \delta > 0$.

Remark 17. The modulus of continuity $\kappa_{X_\bullet}$ is particularly useful as it allows us to promote marginal control on $X_\bullet$ to sample path control. For example, having concentration inequalities on $\kappa_{X_\bullet}$ (c.f., Definition 8) allows us to control the number of crossings of $X_\bullet$ through a low probability region, as seen in Lemma 14. This strategy will be essential in the proofs to come.

Example 18. Concentration inequalities like the one in Definition 8 are typical for stochastic processes used in sampling. For example, consider the classical family of processes

$$X_t = a_t X_0 + b_t X_1 + c_t Z \quad \text{where} \quad (X_0, X_1, Z) \sim P_0 \otimes P_1 \otimes \mathcal{N}(0, 1),$$

where $a, b, c \in C^{0,\eta}([0, 1]; \mathbb{R}^d)$ are Hölder continuous with exponent $0 < \eta < 1$ satisfying appropriate boundary conditions. Now assume that P_0 and P_1 have α - and β -polynomial tails respectively and recall that $\mathcal{N}(0, 1)$ has γ -polynomial tails for every $\gamma > 0$. Since, for any $f \in C^{0,\eta}([0, 1]; \mathbb{R}^d)$, one has

$$\kappa_f(\delta) \leq \delta^\eta \|f\|_{C^{0,\eta}},$$

we may set $M = \|\alpha\|_{C^{0,\eta}} \vee \|\beta\|_{C^{0,\eta}} \vee \|\gamma\|_{C^{0,\eta}}$ and obtain

$$\begin{aligned} \kappa_{X_\bullet}(\delta) &\leq \delta^\eta (\|\alpha\|_{C^{0,\eta}} |X_0| + \|\beta\|_{C^{0,\eta}} |X_1| + \|\gamma\|_{C^{0,\eta}} |Z|) \\ &\leq \delta^\eta M (|X_0| + |X_1| + |Z|) \end{aligned}$$

which by a union bound leads to

$$\begin{aligned} \mathbb{P}(\kappa_{X_\bullet}(\delta) \geq \theta) &\leq \mathbb{P}\left(|X_0| \geq \frac{\theta}{3\delta^\eta M}\right) + \mathbb{P}\left(|X_1| \geq \frac{\theta}{3\delta^\eta M}\right) + \mathbb{P}\left(|Z| \geq \frac{\theta}{3\delta^\eta M}\right) \\ &\lesssim \left(\frac{\theta}{\delta^\eta}\right)^{-\alpha} + \left(\frac{\theta}{\delta^\eta}\right)^{-\beta} + \left(\frac{\theta}{\delta^\eta}\right)^{-\gamma} \\ &\lesssim \left(\frac{\delta^\eta}{\theta}\right)^{\alpha \wedge \beta \wedge \gamma} \\ &= \frac{\delta^{\eta(\alpha \wedge \beta \wedge \gamma)}}{\theta^{\alpha \wedge \beta \wedge \gamma}}. \end{aligned}$$

Thus, with $\tilde{\alpha} := \eta(\alpha \wedge \beta \wedge \gamma)$ and $\tilde{\beta} := \alpha \wedge \beta \wedge \gamma$ and if A is the numerical constant suppressed in the above inequalities, we have shown that

$$X_\bullet \in \mathcal{C}(A, \tilde{\alpha}, \tilde{\beta}).$$

Definition 9. For any real numbers $a < b$ and any stochastic process $X_\bullet : [0, 1] \rightarrow \mathbb{R}$ with continuous sample paths we define the *up-crossing number* of $X_\bullet$ with respect to the interval (a, b) to be the random variable with realizations

$$N_{X_\bullet}(a, b; \omega) = \sup \left\{ n \in \mathbb{N} : \exists t_1 < s_1 < t_2 < s_2 < \dots < t_n < s_n \in [0, 1] \right. \\ \left. \text{such that } X_{t_i}(\omega) \leq a < b \leq X_{s_i}(\omega) \text{ for each } i = 1, \dots, n \right\}.$$

Intuitively, this number $N_{X_\bullet}$ is the number of times a specific realization of the stochastic process crosses through the interval $[a, b]$.

Remark 19. The above definition can easily be generalized to a time dependent interval (a_t, b_t) . Specifically, if $t \mapsto a_t$ and $t \mapsto b_t$ are continuous functions such that $a_t < b_t$ for all $t \in [0, 1]$, then we can define $N_{X_\bullet}(a_\bullet, b_\bullet)$ by replacing the condition $X_{t_i}(\omega) \leq a < b \leq X_{s_i}(\omega)$ with $X_{t_i}(\omega) \leq a_{t_i} < b_{t_i} \leq X_{s_i}(\omega)$ in the above definition.

Lemma 14. Let $a < b$ be real numbers defining an interval $I := (a, b) \subseteq \mathbb{R}$ and let $X_\bullet$ be a stochastic process with continuous sample paths such that

$$\mathbb{P}(X_t \in I) \leq \epsilon \text{ for each } t \in [0, 1].$$

Moreover, suppose there exists $\alpha, \beta > 0$ such that, for all $\theta, \delta > 0$, we have the concentration inequality

$$\mathbb{P}(\kappa_{X_\bullet}(\delta) \geq \theta) \lesssim \frac{\delta^\alpha}{\theta^\beta}. \quad (19)$$

Then, we have

$$\mathbb{P}(N_{X_\bullet}(a, b) \geq 1) \lesssim \left(\frac{\epsilon^\alpha}{(b-a)^\beta} \right)^{\frac{1}{\alpha+1}},$$

where the constant depends on α .

Proof. Start by considering a mesh, i.e, a finite subset $\Pi \subseteq [0, 1]$ consisting of points $t_i = i\delta$ for $i = 0, \dots, [1/\delta]$ with $t_{[1/\delta]+1} = 1$, and note the event inclusion

$$\{N_{X_\bullet}(a, b) \geq 1\} \subseteq \left\{ \exists t_i \in \Pi : X_{t_i} \in I \right\} \cup \left\{ \kappa_{X_\bullet}(\delta) \geq b - a \right\}. \quad (20)$$

In other words, trajectories $X_\bullet$ that cross through (a, b) at least once must either have at least one mesh point t_i such that $X_{t_i} \in (a, b)$ or move through (a, b) in a time interval smaller than the mesh size δ . Indeed, fix a realization $\omega \in \Omega$ and suppose that $N_{X_\bullet}(a, b; \omega) \geq 1$. This implies that there exists $s < t \in [0, 1]$ such that

$$X_s(\omega) \leq a < b \leq X_t(\omega).$$

Now, if there exists $t_i \in \Pi$ with $s \leq t_i \leq t$, then it follows that $X_{t_i}(\omega) \in I$ by continuity of the sample path $t \mapsto X_t(\omega)$. On the other hand, if no such t_i exists then

$$|s - t| \leq \delta.$$

In particular, the definition of the modulus κ gives

$$\kappa_{X_\bullet}(\delta; \omega) \geq |X_t(\omega) - X_s(\omega)| \geq b - a,$$

which completes the proof of the event inclusion [\(20\)](#).

Now applying a union bound to the right-hand side of equation [\(20\)](#), we obtain

$$\begin{aligned}\mathbb{P}(\exists t \in [0, 1], X_t \in S) &\leq \sum_{t_i \in \Pi} \mathbb{P}(X_{t_i} \in S) + \mathbb{P}(\kappa_{X_\bullet}(\delta) \geq (b-a)) \\ &\lesssim \frac{1}{\delta} \epsilon + \frac{\delta^\alpha}{(b-a)^\beta}.\end{aligned}$$

Minimizing the auxiliary function

$$f(\delta) = \frac{1}{\delta} \epsilon + \frac{\delta^\alpha}{(b-a)^\beta}$$

we obtain the minimizer

$$\delta^\star = \left(\frac{(b-a)^\beta}{\alpha} \epsilon \right)^{\frac{1}{\alpha+1}},$$

with minimum value

$$f(\delta^\star) = \left(\frac{\alpha+1}{\alpha^{\frac{\alpha}{\alpha+1}}} \right) \frac{\epsilon^{\frac{\alpha}{\alpha+1}}}{(b-a)^{\frac{\beta}{\alpha+1}}}.$$

Plugging $\delta^\star$ in the above bound for the event $\{X_t \in S\}$ yields the desired result. $\square$

Corollary 15. *Let $a, b : [0, 1] \rightarrow \mathbb{R}$ be continuous functions such that $a_t < b_t$ for all $t \in [0, 1]$, defining the time dependent interval $I_t = (a_t, b_t)$. Assume that there exists a real number $\Delta I > 0$ such that $b_t - a_t \geq \Delta I$ for all $t \in [0, 1]$. Then, under the assumptions of Lemma [14](#) we have*

$$\mathbb{P}(N_{X_\bullet}(a_\bullet, b_\bullet) \geq 1) \lesssim \left(\frac{\epsilon^\alpha}{\Delta I^\beta} \right)^{\frac{1}{\alpha+1}},$$

with the constant depending on α .

Proof. The proof follows Lemma [14](#) mutatis mutandis. $\square$

Remark 20. We note that choosing $[0, 1]$ for the domain of the process $X_\bullet$ is immaterial and the proof would have gone through verbatim for any $[s_0, s_1]$ with $s_0 < s_1 \in \mathbb{R}$.

The technology we have developed so far allows us to answer a question posed in the end of the last subsection: namely, was the result of Theorem [11](#) an artifact of the disjoint supports of the target measure? The answer is *no*, as one can see from the following result.

Proposition 16. *For all $A, \alpha, \beta > 0$ there exist $\mu, \sigma > 0$ such that the mixture model*

$$P = \frac{1}{2} \mathcal{N}(\mu, \sigma) + \frac{1}{2} \mathcal{N}(-\mu, \sigma)$$

satisfies

$$\mathcal{S}_{\text{SL}}(P, P) \cap \mathcal{C}(A, \alpha, \beta) = \emptyset.$$

Proof sketch. With an appropriate modification of the second step of the proof of Theorem [11](#) one can show that for any process $X_\bullet \in \mathcal{S}_{\text{SL}}(P, P)$ there exists $a \in \mathbb{R}$ with $0 < a < \mu$ such that the interval $I = (-a, a)$ satisfies

$$\mathbb{P}(X_t \in I) \leq \epsilon, \quad \forall t \in [0, 1],$$

A direct application of Lemma [14](#) then yields that

$$\mathbb{P}\left(N_{X_\bullet}(-a, a) \geq 1\right) \leq A \left(\frac{\epsilon^\alpha}{(2a)^\beta}\right)^{\frac{1}{\alpha+1}},$$

and the right-hand side can be made arbitrarily small by taking $\epsilon \searrow 0$ which is possible by taking $\sigma \searrow 0$ and keeping μ fixed. On the other hand, we have

$$\mathbb{P}\left(X_0 < -\mu \text{ and } X_1 > \mu\right) = \frac{1}{16},$$

and then by definition of an up-crossing we get

$$\mathbb{P}\left(N_{X_\bullet}(-a, a) \geq 1\right) \geq \mathbb{P}\left(X_0 < -\mu \text{ and } X_1 > \mu\right) = \frac{1}{16},$$

yielding a contradiction and completing the proof. $\square$

We can now state and prove the main result of this section, which generalizes Theorem [11](#) in two ways. First, the target measure has connected support—in fact, it can also have full support, as seen in Proposition [16](#). Second, the source measure now includes *all* positive absolutely continuous measures satisfying some mild regularity assumptions. In particular, the result holds for a standard normal source measure, which is commonly used in practice. Before stating and proving the result, we make the needed regularity of the source measure precise.

Definition 10. A measure $P \in \mathcal{P}(\mathbb{R}^d)$ is *positive absolutely continuous* if it is absolutely continuous with respect to the Lebesgue measure and, for any Borel set $A \subseteq \mathbb{R}$ with positive Lebesgue measure, we have $P(A) > 0$.

Definition 11. A measure $P \in \mathcal{P}(\mathbb{R}^d)$ is said to be (C, γ) -Frostman for some $C, \gamma > 0$ if we have

$$P(B_r(x)) \leq C r^\gamma,$$

for all $x \in \mathbb{R}^d$ and $r > 0$, where $B_r(x) = \{y \in \mathbb{R}^d : \|y - x\| < r\}$ is an open ball.

Example 21. Frostman measures are well-studied in the literature, see for example [\[Mat12 Theorem 8.8\]](#). In fact, the class of Frostman measures contains absolutely continuous measures with $L^p(\mathbb{R}^d)$ densities for any $p \in (1, \infty]$. Indeed, for $p = \infty$ and a measure P with density $\pi \in L^\infty(\mathbb{R}^d)$ we have

$$P(B_r(x)) = \int_{B_r(x)} \pi(y) dy \leq \|\pi\|_\infty |B_r(x)| = \|\pi\|_\infty |B_1(0)| r^d,$$

where $|B_r(x)|$ is the Lebesgue measure of the ball $B_r(x)$. Thus, measure P is (C, d) -Frostman with $C = \|\pi\|_\infty |B_1(0)|$. For $p \in (1, \infty)$ we can apply Hölder's inequality to obtain

$$P(B_r(x)) = \int_{B_r(x)} \pi(y) dy \leq \|\pi\|_p |B_r(x)|^{1-\frac{1}{p}} = \|\pi\|_p |B_1(0)|^{1-\frac{1}{p}} r^{\frac{d(p-1)}{p}},$$

thus P is $(C, \frac{d(p-1)}{p})$ -Frostman with $C = \|\pi\|_p |B_1(0)|^{1-\frac{1}{p}}$.

Theorem 17. Let $P_0 \in \mathcal{P}(\mathbb{R})$ be any positive absolutely continuous measure that is (C, γ) -Frostman for some $C, \gamma > 0$. Moreover, let P_1 be any absolutely continuous measure with properly ϵ -disconnected support for some $\epsilon > 0$ and positive weights $w_i := P_1(S_i) > 0$ for $i = 0, 1$. Then, for any real numbers $A, \alpha, \beta > 0$ there exists an $\epsilon_0 = \epsilon_0(C, \gamma, S_0, S_1, w_0, w_1, A, \alpha, \beta)$ depending only on the numbers C, γ , the support geometries S_0, S_1 , the real numbers w_0, w_1 and the numbers A, α, β such that if $\epsilon \leq \epsilon_0$ then we have

$$\mathcal{S}_{\text{SL}}(P_0, P_1) \cap \mathcal{C}(A, \alpha, \beta) = \emptyset.$$

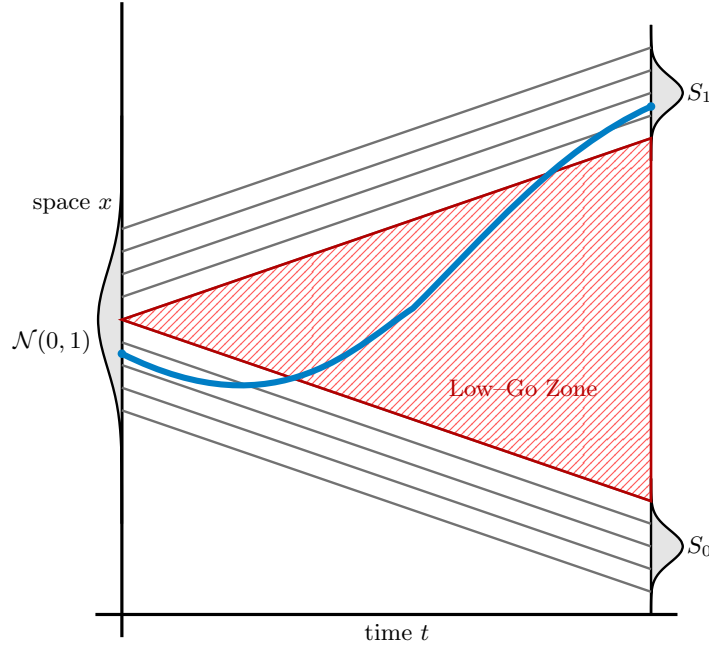


Figure 5: Schematic depiction of the proof of Theorem 17

Proof of Theorem 17. Let S_0, S_1 be as in Definition 5 with $d = 1$. By shifting the origin if necessary, we can assume without loss of generality that

$$\sup S_0 = s_0 < 0 < s_1 := \inf S_1. \quad (21)$$

Moreover, sample $X \sim P_0$ and let $x_0, x_1 \in \mathbb{R}$ denote the unique real numbers satisfying

$$\mathbb{P}(X \leq x_0) = P_1(S_0) \quad \text{and} \quad \mathbb{P}(X \geq x_1) = P_1(S_1).$$

We know these are unique by the construction of P_0 as strictly positive and absolutely continuous. By the fact that $P_1(S_0) + P_1(S_1) \geq 1 - \epsilon > 0$ it follows that

$$\mathbb{P}(x_0 \leq X \leq x_1) \leq \epsilon.$$

Working towards a contradiction, suppose there exists

$$X_\bullet \in \mathcal{S}_{\text{SL}}(P_0, P_1) \cap \mathcal{C}(\alpha, \beta).$$

First, note that **Step 1** in the proof of Theorem [11](#) extends verbatim. We proceed by modifying **Step 2** to **Step 4**.

Step 2: Topological Obstruction

For $x, y \in \mathbb{R}$ define the space-time line-segment that connects the points $(0, x)$ and $(1, y)$ by

$$\ell(x, y) := \left\{ (t, z) \in [0, 1] \times \mathbb{R} : z = (y - x)t + x \right\},$$

Recall that the epigraph and hypograph of a function $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ are defined as

$$\text{epi}[f] = \{(x, y) \in D \times \mathbb{R} : y \geq f(x)\} \quad \text{and} \quad \text{hypo}[f] = \{(x, y) \in D \times \mathbb{R} : y \leq f(x)\}.$$

We can now construct the space-time half-spaces

$$H_+ := \text{hypo}[\ell(x_0, s_0)] \quad \text{and} \quad H_- := \text{epi}[\ell(x_1, s_1)].$$

Now, consider the augmented flow map

$$\Phi : [0, 1] \times \mathbb{R} \rightarrow [0, 1] \times \mathbb{R}, \quad \Phi(t, x) = (t, \phi(t, x)).$$

If we define the half-lines

$$I_+ := \{x \in \mathbb{R} : x \geq x_1\}, \quad I_- := \{x \in \mathbb{R} : x \leq x_0\}.$$

then the goal of this step is to show that for all $t \in [0, 1]$ we have

$$\Phi(t, I_-) \subseteq H_- \quad \text{and} \quad \Phi(t, I_+) \subseteq H_+. \quad (22)$$

By the global injectivity of the flow, established in **Step 1** of Theorem [11](#) we observe that it suffices to show that

$$\phi(1, x_i) = s_i, \quad i = 0, 1. \quad (23)$$

Indeed, if [\(23\)](#) is satisfied but [\(22\)](#) is violated then there exists, e.g., coordinate $(t, x) \in [0, 1] \times I_-$ such that the augmented flow map lies in inappropriate half-space, i.e., $\Phi(t, x) \in H_-^c$. This means the flow map $\phi(t, x) \in \mathbb{R}$ is above the line $\ell(x_0, s_0)$ at time t . Since $x < x_0$ and the curve $t \mapsto \phi(t, x)$ is continuous, it must cross the line $\ell(x_0, s_0)$ at some time $s < t$ yielding $\phi(s, x) = \phi(s, x_0)$ thus violating injectivity.

We now establish the validity of [\(23\)](#). Consider first the case $i = 0$ and suppose, for the purpose of contradiction, that $\phi(1, x_0) < s_0$. By the injectivity and continuity of the flow (c.f., the above paragraph) it follows that $\phi(1, \cdot)$ is monotone: it must be that $\phi(1, x) < s_0$ for all $x < x_0$ and similarly $\phi(1, x) > s_0$ for all $x > x_0$. Therefore

$$\begin{aligned} \mathbb{P}(\phi(1, X_0) \leq \phi(1, x_0)) &= \mathbb{P}(X_0 \leq x_0) \\ &= P_0((-\infty, x_0]) \\ &= P_1(S_0), \end{aligned} \quad (24)$$

where in the last step we have used the choice of x_0 . By Corollary [26](#) however, we have

$$X_t \stackrel{d}{=} \phi(t, X_0), \quad \text{for all } t \in [0, 1],$$

for $X_\bullet$ the process inducing the flow ϕ . Since $X_1 \sim P_1$ we obtain

$$\begin{aligned} \mathbb{P}(\phi(1, X_0) \leq \phi(1, x_0)) &= \mathbb{P}(X_1 \leq \phi(1, x_0)) \\ &= \mathbb{P}(X_1 \leq s_0) - \mathbb{P}(\phi(1, x_0) < X_1 \leq s_0) \\ &< P_1(S_0), \end{aligned}$$

and in the last step we have used that properness of the support of P_1 , c.f. Definition [6](#) implying that P_1 is positive on the interval $(\phi(1, x_0), s_0)$. This yields a contradiction with [24](#).

On the other hand, consider instead assuming $\phi(1, x_0) > s_0$ for contradiction. Then, by the continuity of the map $x \mapsto \phi(1, x)$, there is a real number $h > 0$ such that, if $y \in (x_0 - h, x_0 + h)$, then $\phi(1, y) > s_0$. By the monotonicity of $\phi(1, \cdot)$ discussed above, we can further conclude that having $y \in (x_0 - h, x_0)$ implies property $\phi(1, x) > \phi(1, y) > s_0$. Thus, we have

$$\begin{aligned} \mathbb{P}(\phi(1, X_0) \leq \phi(1, x_0)) &= \mathbb{P}(\phi(1, X_0) \leq s_0) - \mathbb{P}(s_0 \leq \phi(1, X_0) \leq \phi(1, x_0)) \\ &\leq \mathbb{P}(\phi(1, X_0) \leq s_0) - \mathbb{P}(x_0 - h \leq X_0 \leq x_0) \\ &< \mathbb{P}(\phi(1, X_0) \leq s_0), \end{aligned}$$

using positivity of P_0 in the last step. Recall that Corollary [26](#) gives $\phi(1, \cdot)_\# P_0 = P_1$, and so the last inequality can be rewritten as

$$\mathbb{P}(\phi(1, X_0) \leq \phi(1, x_0)) < \mathbb{P}(X_1 \leq s_0) = P_1(S_0).$$

The monotonicity of $\phi(1, \cdot)$, however, implies that

$$\mathbb{P}(\phi(1, X_0) \leq \phi(1, x_0)) = \mathbb{P}(X_0 \leq x_0) = P_1(S_0),$$

by the choice of x_0 , yielding a contradiction and completing the proof of [23](#) for $i = 0$. The case $i = 1$ is handled similarly. This completes the proof of [22](#).

Step 3: Low-go zone

Define the *low-go zone* and its time-slices, for each $t \in [0, 1]$, by

$$G = (H_+ \cup H_-)^c \quad \text{and} \quad G_t = \{x \in \mathbb{R} : (t, x) \in G\}.$$

Recall that Corollary [26](#) gives us the distributional equality

$$X_t \stackrel{d}{=} \phi(t, X_0), \quad \text{for all } t \in [0, 1]. \tag{25}$$

Now, we have

$$\begin{aligned}
\mathbb{P}(X_t \in G_t) &= \mathbb{P}(\phi(t, X_0) \in G_t) \\
&= \mathbb{P}(\Phi(t, X_0) \in G) \\
&= \mathbb{P}(\Phi(t, X_0) \in (H_+ \cup H_-)^c) \\
&\leq \mathbb{P}(X_0 \in (I_- \cup I_+)^c)
\end{aligned}$$

using equality (25) in the first line, the definition of Φ in the second line, the definition of G in the third line and (22) in the fourth line. Recalling the definition of I_- and I_+ we have

$$(I_- \cup I_+)^c = I_-^c \cap I_+^c = \{x \in \mathbb{R} : x_0 < x < x_1\},$$

and recalling the definition of x_0 and x_1 we get

$$(I_- \cup I_+)^c = (S_0 \cup S_1)^c.$$

Since S_0 and S_1 are as in Definition 5 we have that $P_0(S_0 \cup S_1) \geq 1 - \epsilon$ and so

$$\mathbb{P}(X_t \in G_t) \leq \epsilon, \quad \text{for all } t \in [0, 1]. \quad (26)$$

Step 4: Small probability of crossing

Since H_- and H_+ are disconnected and closed, any continuous curve that starts in H_- and ends in H_+ must cross $G = (H_- \cup H_+)^c$. For $t \leq t' \in [0, 1]$ consider the event of crossing the low-go zone G between times t and t' , i.e.,

$$C_{t,t'} = \left\{ \exists u_+, u_- \in [t, t'] : (u_+, X_{u_+}) \in H_+, (u_-, X_{u_-}) \in H_- \right\},$$

In this step, we use (26) to construct a function $g : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $g(\epsilon) \searrow 0$ as $\epsilon \searrow 0$ and

$$\mathbb{P}(C_{0,1}) = g(\epsilon). \quad (27)$$

Let $\delta > 0$ be a parameter to be chosen later. By a union bound, we have that

$$\mathbb{P}(C_{0,1}) \leq \mathbb{P}(C_{0,\delta}) + \mathbb{P}(C_{\delta,1}) + \mathbb{P}(X_\delta \in G_\delta), \quad (28)$$

where we have used that any crossing that starts in $(0, \delta)$ and ends in $(\delta, 1)$ must be in the low-go zone G_δ at time δ . Now, the inequality in (26) allows us to immediately control the last term in (28), yielding

$$\mathbb{P}(X_\delta \in G_\delta) \leq \epsilon. \quad (29)$$

For the second term, we argue as follows: recall the definition of G and that of H_- and H_+ as the hypograph and epigraph of the line-segments $\ell(x_0, s_0)$ and $\ell(x_1, s_1)$ respectively. We now see that G is a space-time trapezoid, defined by the non-intersecting line-segments $\ell(x_0, s_0)$ and $\ell(x_1, s_1)$ as well as the parallel lines $\{t(0, x_0 - x_1) : t \in [0, 1]\}$ and $\{t(1, s_0 - s_1) : t \in [0, 1]\}$. Therefore, we can conclude that for any $t \in [0, 1]$ there are real numbers $a_t < b_t$ such that

$$G_t = \{x \in \mathbb{R} : a_t < x < b_t\}.$$

For a fixed $\epsilon > 0$ we can distinguish two cases:

Case 1: $|x_0 - x_1| < \frac{1}{2}|s_0 - s_1|$

Here, the line-segments $\ell(x_0, s_0)$ and $\ell(x_1, s_1)$ diverge as t increases, yielding the estimate

$$|G_t| \geq |G_\delta| \geq \mu \delta, \quad \text{for all } t \in [\delta, 1],$$

with

$$\mu = |s_1 - s_0 - (x_1 - x_0)| > \frac{|s_1 - s_0|}{2} > 0,$$

using the reverse triangle inequality. Finally, using Lemma 14, we obtain

$$\begin{aligned} \mathbb{P}(C_{\delta,1}) &\leq \mathbb{P}(N_{X_\bullet}(a_\delta, b_\delta) \geq 1) \\ &\leq A \left(\frac{\epsilon^\alpha}{|G_\delta|^\beta} \right)^{\frac{1}{\alpha+1}}, \end{aligned}$$

and, using the above lower bounds on $|G_\delta|$ and the constant μ , we obtain

$$\mathbb{P}(C_{\delta,1}) \leq \frac{2^{\frac{\beta}{\alpha+1}} A}{|s_0 - s_1|^{\frac{\beta}{\alpha+1}}} \left(\frac{\epsilon^\alpha}{\delta^\beta} \right)^{\frac{1}{\alpha+1}}.$$

Case 2: $|x_0 - x_1| \geq \frac{1}{2}|s_0 - s_1|$

Here, the line-segments $\ell(x_0, s_0)$ and $\ell(x_1, s_1)$ converge as t increases, so we have the estimate

$$|G_t| \geq |G_1| = \frac{|s_0 - s_1|}{2}, \quad \text{for all } t \in [\delta, 1],$$

Therefore, using Lemma 14 we obtain

$$\mathbb{P}(C_{\delta,1}) \leq A \left(\frac{\epsilon^\alpha}{|G_1|^\beta} \right)^{\frac{1}{\alpha+1}} = \frac{2^{\frac{\beta}{\alpha+1}} A}{|s_0 - s_1|^{\frac{\beta}{\alpha+1}}} \epsilon^{\frac{\alpha}{\alpha+1}}.$$

Therefore, we can combine the above two cases to obtain the following upper bound on the second term in the union bound (28):

$$\mathbb{P}(C_{\delta,1}) \leq \frac{2^{\frac{\beta}{\alpha+1}} A}{|s_0 - s_1|^{\frac{\beta}{\alpha+1}}} \left(\left(\frac{\epsilon^\alpha}{\delta^\beta} \right)^{\frac{1}{\alpha+1}} + \epsilon^{\frac{\alpha}{\alpha+1}} \right). \quad (30)$$

We now turn our attention to the first term $\mathbb{P}(C_{0,\delta})$. We shall now control this term as $\delta \rightarrow 0$ ⁴. Introduce a parameter $\xi > 0$ to be chosen later and decompose the event $C_{0,\delta}$ into the following cover:

$$C_{0,\delta}^{\xi-} := C_{0,\delta} \cap \left\{ \omega : \left| X_0(\omega) - \frac{x_0 + x_1}{2} \right| \leq \xi \right\}, \quad C_{0,\delta}^{\xi+} := C_{0,\delta} \cap \left\{ \omega : \left| X_0(\omega) - \frac{x_0 + x_1}{2} \right| > \xi \right\}.$$

⁴Note that upper continuity of the measure $\mathbb{P}$ implies that

$$\lim_{\delta \rightarrow 0} \mathbb{P}(C_{0,\delta}) = \mathbb{P}\left(\bigcap_{\delta > 0} C_{0,\delta}\right) = \mathbb{P}(C_{0,0}) = 0, \quad (31)$$

and the last step follows by the definition of the event $C_{0,0}$. However, to proceed we need *quantitative control* on the rate at which $\mathbb{P}(C_{0,\delta})$ goes to zero as $\delta \rightarrow 0$.

Since the events $C_{0,\delta}^{\xi_-}$ and $C_{0,\delta}^{\xi_+}$ form a partition of $C_{0,\delta}$ we have

$$\mathbb{P}(C_{0,\delta}) = \mathbb{P}(C_{0,\delta}^{\xi_-}) + \mathbb{P}(C_{0,\delta}^{\xi_+})$$

and using the (C, γ) -Frostman assumption on P_0 we have

$$\mathbb{P}(C_{0,\delta}^{\xi_-}) \leq \mathbb{P}\left(\left|X_0 - \frac{x_0 + x_1}{2}\right| \leq \xi\right) \leq C \xi^\gamma.$$

On the other hand, we can use that $X_\bullet \in \mathcal{C}(A, \alpha, \beta)$ and Corollary [15](#) to obtain

$$\begin{aligned} \mathbb{P}(C_{0,\delta}^{\xi_+}) &\leq \mathbb{P}\left(N_{X_\bullet}\left(\frac{x_0 + x_1}{2} - \xi, \frac{x_0 + x_1}{2}\right) \geq 1\right) + \mathbb{P}\left(N_{X_\bullet}\left(\frac{x_0 + x_1}{2}, \frac{x_0 + x_1}{2} + \xi\right) \geq 1\right) \\ &\leq 2A \left(\frac{\delta^\alpha}{\xi^\beta}\right)^{\frac{1}{\alpha+1}}. \end{aligned}$$

Therefore, we obtain the upper bound

$$\mathbb{P}(C_{0,\delta}) \leq C \xi^\gamma + 2A \frac{\delta^{\frac{\alpha}{\alpha+1}}}{\xi^{\frac{\beta}{\alpha+1}}},$$

and minimizing the right-hand side over $\xi > 0$ we obtain

$$\mathbb{P}(C_{0,\delta}) \leq B(\alpha, \beta, \gamma, A, C) \delta^{\frac{\alpha\gamma}{\gamma(\alpha+1)+\beta}},$$

for a constant $B(\alpha, \beta, \gamma, A, C) > 0$ that depends only on the parameters α, β, γ, A and C [5](#). We can now select

$$\delta(\epsilon) = \epsilon^{\frac{\alpha}{2\beta}},$$

and combine equation [\(29\)](#), equation [\(30\)](#) and equation [\(31\)](#) in the union bound [\(28\)](#) to obtain

$$\mathbb{P}(C_{0,1}) \leq D(\alpha, \beta, \gamma, A, C, S_0, S_1) \left(\epsilon^{\frac{\alpha^2\gamma}{2\beta\gamma(\alpha+1)+2\beta^2}} + \epsilon^{\frac{\alpha}{2(\alpha+1)}} + \epsilon^{\frac{\alpha}{\alpha+1}} + \epsilon \right). \quad (32)$$

and $D(\alpha, \beta, \gamma, A, C, S_0, S_1) > 0$ is a constant that depends only on the parameters $\alpha, \beta, \gamma, A, C$ and the sets S_0 and S_1 . We have, therefore, completed the proof of the asserted equality [\(27\)](#) with the function g defined as the right-hand side of the above display.

Step 5: Contradiction

To conclude the proof, we note that by independence we have

$$\mathbb{P}(X_0 \leq x_0 \text{ and } X_1 \in S_1) = \mathbb{P}(X_0 \leq x_0) \mathbb{P}(X_1 \in S_1) = P_1(S_0) P_1(S_1),$$

⁵In fact, we have the explicit expression

$$B(\alpha, \beta, \gamma, A, C) = \left[\left(\frac{\frac{\beta}{\alpha} + 1}{\gamma} \right)^{\frac{\gamma}{\gamma + \frac{\beta}{\alpha+1}}} + \left(\frac{\gamma}{\frac{\beta}{\alpha+1}} \right)^{\frac{\frac{\beta}{\alpha+1}}{\gamma + \frac{\beta}{\alpha+1}}} \right] C^{\frac{\frac{\beta}{\alpha+1}}{\gamma + \frac{\beta}{\alpha+1}}} (2A)^{\frac{\gamma}{\gamma + \frac{\beta}{\alpha+1}}}.$$

On the other hand, we have the event inclusions

$$\{X_0 \leq x_0\} \subseteq \{(0, X_0) \in H_-\} \quad \text{and} \quad \{X_1 \in S_1\} \subseteq \{(1, X_1) \in H_+\},$$

therefore, we can write

$$\begin{aligned} \mathbb{P}(X_0 \leq x_0 \text{ and } X_1 \in S_1) &\leq \mathbb{P}((0, X_0) \in H_- \text{ and } (1, X_1) \in H_+) \\ &\leq \mathbb{P}(C_{0,1}). \end{aligned}$$

Set $w_i := P_1(S_i)$ for $i = 0, 1$ as in the statement of this theorem. Using equation (32) and the above displays we obtain the bound

$$w_0 w_1 \leq D(\alpha, \beta, \gamma, A, C, S_0, S_1) \left(\epsilon^{\frac{\alpha^2 \gamma}{2\beta \gamma(\alpha+1) + 2\beta^2}} + \epsilon^{\frac{\alpha}{2(\alpha+1)}} + \epsilon^{\frac{\alpha}{\alpha+1}} + \epsilon \right), \quad (33)$$

Since all exponents of ϵ in the right-hand side are positive the right-hand side a decreasing function of ϵ that goes to zero as $\epsilon \searrow 0$.

Therefore, there exists an $\epsilon_0 = \epsilon_0(\alpha, \beta, \gamma, A, C, S_0, S_1) > 0$ such that

$$w_0 w_1 > D(\alpha, \beta, \gamma, A, C, S_0, S_1) \left(\epsilon_0^{\frac{\alpha^2 \gamma}{2\beta \gamma(\alpha+1) + 2\beta^2}} + \epsilon_0^{\frac{\alpha}{2(\alpha+1)}} + \epsilon_0^{\frac{\alpha}{\alpha+1}} + \epsilon_0 \right).$$

Clearly, for any $\epsilon \leq \epsilon_0$ the necessary inequality (33) is violated, yielding a contradiction and completing the proof of the theorem. $\square$

Remark 22. We point out that an essential ingredient to the proof above was the scaling

$$|G_\delta| \asymp \delta,$$

in the case $|x_0 - x_1| < \frac{1}{2}|s_0 - s_1|$ —this is a consequence of the geometry of the low-go zone G describing a trapezoidal space-time region. Under the same assumption $|x_0 - x_1| < \frac{1}{2}|s_0 - s_1|$, if one tried to *erroneously* do the above proof by restricting their attention to the smaller rectangular space-time region

$$I(\epsilon) = [0, 1] \times [x_0(\epsilon), x_1(\epsilon)] \subsetneq G,$$

recalling the explicit dependence of $x_i = x_i(\epsilon)$ for $i = 0, 1$, see the beginning of the proof above. Suppose for concreteness that P_0 is a standard normal distribution and that $S_0 = -S_1$ are symmetric around zero. Then, $x_0(\epsilon) = -x_1(\epsilon)$ and $x_0(\epsilon)$ is the unique solution to the equation

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x_0(\epsilon)} e^{-\frac{x^2}{2}} dx = \frac{1 - \epsilon}{2}. \quad (34)$$

By Lemma (14) we have

$$\mathbb{P}(N_{X_\bullet}(-x_0(\epsilon), x_0(\epsilon))) \leq A \left(\frac{\epsilon^\alpha}{2x_0^\beta(\epsilon)} \right)^{\frac{1}{\alpha+1}}, \quad (35)$$

and using equation (34) we can straightforwardly compute that

$$\lim_{\epsilon \downarrow 0} \frac{x_0(\epsilon)}{\epsilon} = c \neq 0.$$

Thus, for the right hand side of (35) to vanish as $\epsilon \searrow 0$, a *necessary* ingredient of our proof strategy, it must be that $\alpha > \beta$. Is this reasonable? Looking at Example 18 we see that to enforce this for a typical sampling processes of the form $X_t = a_t X_0 + b_t X_1 + c_t Z$ we would need to take $a, b, c \in C^{0,\eta}([0, 1]; \mathbb{R}^d)$ with $\eta > 1$. A classical argument shows that on connected domains $\eta > 1$ implies that a, b, c are constant functions, which is impossible since, in particular, we want $a_0 = 1$ and $a_1 = 0$. We conclude that the class $\mathcal{C}(A, \alpha, \beta)$ with $\alpha > \beta$ is somewhat trivial, since it contains none of the processes we typically use in sampling.

Corollary 18. *Fix $P_0 = \mathcal{N}(0, 1)$ and weights $w_0, w_1 > 0$ with $w_0 + w_1 = 1$ and positive reals $A, \alpha, \beta > 0$ defining the class $\mathcal{C}(A, \alpha, \beta)$. There exist $\mu_i \in \mathbb{R}$ and $\sigma_i > 0$ for $i \in \{0, 1\}$, such that*

$$P_1 = w_0 \mathcal{N}(\mu_0, \sigma_0) + w_1 \mathcal{N}(\mu_1, \sigma_1)$$

satisfies

$$\mathcal{S}_{\text{SL}}(P_0, P_1) \cap \mathcal{C}(A, \alpha, \beta) = \emptyset.$$

Proof. Note that P_0 is a (C, γ) -Frostman measure for explicit constants $C, \gamma > 0$ since it has a bounded density, see Example 21. Now we can select $\mu_i \in \mathbb{R}$ such that the sets $S_0 = \{x \in \mathbb{R} : |x - \mu_0| < 1\}$ and $S_1 = \{x \in \mathbb{R} : |x - \mu_1| < 1\}$ are disjoint. Now for any $\epsilon > 0$, there exist $\sigma_0, \sigma_1 > 0$ such that the measure P_1 defined as above has properly ϵ -disconnected support on the sets S_0 and S_1 , as per Definition 6. By Theorem 17 there exists $\epsilon_0 = \epsilon_0(C, \gamma, S_0, S_1, w_0, w_1, A, \alpha, \beta) > 0$ such that if $\epsilon < \epsilon_0$ then we have

$$\mathcal{S}_{\text{SL}}(P_0, P_1) \cap \mathcal{C}(A, \alpha, \beta) = \emptyset.$$

In particular, we can select $\sigma_0, \sigma_1 > 0$ such that $\epsilon = \frac{\epsilon_0}{2} < \epsilon_0$ thereby completing the proof. $\square$

5 Conclusion & Discussion

In this work, we show many results regarding the construction and impossibility of straight-line, i.e., one-shot, flows under the independent endpoint constraint $(X_0, X_1) \sim P_0 \otimes P_1$. In particular, we address when a stochastic process $X_\bullet$ induces a flow map with vanishing acceleration in order to perform sampling with one evaluation of the velocity v_t . This idea leads to a structural theory of straightness at the level of stochastic processes rather than the level of any particular algorithmic parameterization. Our first contribution is analytical, as we develop several equivalent characterizations of a straight-line process by linking vanishing flow acceleration, vanishing material derivative $D_t v_t = 0$ for the conditional velocity field, and a balance law described by a PDE. These characterizations give a strong analytical handle on candidate processes, distinguishing the conditional velocity v_t and the material derivative D_t as the central objects to a straight flow. We further address the typical affine path $X_t = (1 - t)X_0 + tX_1$, where straightness occurs if and only if the endpoint coupling is deterministic. In this sense, constructing a transport map (e.g., the Monge map) is not just a canonical example, but the unique mechanism for achieving straightness with the affine path.

Our second contribution is constructive, as we prove that endpoint independence does not, by itself, prevent straight-line flows. For arbitrary Gaussian marginals P_0 and P_1 , we construct explicit generalized interpolants inducing straight-line flows and satisfying $(X_0, X_1) \sim P_0 \otimes P_1$. This gives a genuinely nontrivial example: one-shot straight generation is possible without sacrificing the independence of the endpoints (unlike, e.g., using a Cholesky-based transport between the two measures). At the same time, our constructed flows highlight the special role played by the Gaussian structure, as we explicitly compute all relevant conditional quantities.

Our last contribution shows a core limitation of the modern generative flow paradigm. We show that, for a target P_1 with well-separated modes, straightness is fundamentally incompatible with endpoint independence. The obstruction is geometric: straight-line flows impose a rigid space-time geometry on the image of the source measure under the flow. Processes with independent endpoints, however, cannot possibly satisfy such rigid constraints. We establish this fundamental contradiction through a hierarchy of impossibility theorems, beginning with the one-dimensional topological core of this contradiction, then extending to more general source and target measures in any dimension. In doing so, we develop novel analytical techniques for modern transport-based generative modelling, entailing a mixture of geometry, topology, and classical stochastic process theory. Taken together, these results yield two distinct ends of a spectrum: straight line flows with endpoint independence are explicit in highly structured settings, but are impossible in many other settings for reasons that are intrinsic rather than algorithmic.

Open questions. Several natural directions remain open:

1. *Extend the impossibility theory in dimensions $d > 1$.* Our higher-dimensional impossibility results, i.e., Theorem [13](#) are for measures with disjoint support. We believe that using the analytical techniques developed in Theorem [17](#) could allow for such extensions. In $d = 1$ we were able to understand the geometry of the flow explicitly; such a description is not available in $d > 1$ so our techniques need to be extended.
2. *Construct explicit straight-line processes beyond the Gaussian setting.* The Gaussian case is special because the relevant conditional statistics can be computed in closed form. It is therefore natural to ask whether other classes of targets, such as log-concave or more general unimodal distributions, also admit explicit straight independent endpoint constructions. The balance law formulation developed here may offer a useful starting point for this question.
3. *Understand intermediate regimes between independence and determinism.* The present work studies one extreme in which (X_0, X_1) are independent, while the classical optimal-transport construction corresponds to the opposite extreme of a deterministic coupling. It would be very interesting to understand what becomes possible when the coupling is only partial. More generally, one may ask how the degree of endpoint dependence controls the existence of straight or nearly straight flows, and whether this yields a finer theory of the tradeoff between pre-computation and inference-time efficiency.

This work contributes to the theory of flow-based generative modelling by addressing the practical question of inference efficiency. We show the best-case scenario, prescribing straight-line processes for Gaussian endpoints constructively, and we show the worst-case scenario, describing problem instances in which we cannot construct *any* straight-line flow. The ramifications to filling in the spectrum between best- and worst-case are clear: by creating models in which the paths are straight,

we can address major concerns of practicality and computation throughout the paradigm of generative flows.

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A Section 2 proofs

Assumption. For the remainder of this section, assume that the marginal densities ρ_t of X_t exist for all $t \in [0, 1]$ and the sample paths $t \mapsto X_t(\omega)$ are in $W^{2,2}([0, 1]; \mathbb{R}^d)$ for each realization $\omega \in \Omega$. Moreover, assume that the trajectory maps $t \mapsto \phi_t(x)$ are in $C^2([0, 1]; \mathbb{R}^d)$ for each $x \in \mathbb{R}^d$ whereas for each $t \in [0, 1]$ the flow map $x \mapsto \phi_t(x)$ are homeomorphisms. Lastly, assume the conditional velocity $v : (t, x) \mapsto v_t(x)$ is a continuously differentiable map jointly in $[0, 1] \times \mathbb{R}^d$.

Proposition 19. *The following are equivalent:*

1. For all $(t, x) \in [0, 1] \times \mathbb{R}^d$, one has

$$\frac{d^2}{dt^2} \phi_t(x) = 0.$$

2. For all $(t, x) \in [0, 1] \times \mathbb{R}^d$, one has

$$\phi_t(x) = (1 - t)x + t\phi_1(x).$$

3. For all $(t, x) \in [0, 1] \times \mathbb{R}^d$, one has

$$D_t v_t(x) = 0.$$

Proof. It is clear that (2) $\implies$ (1). Let us show that (1) $\implies$ (2). We have

$$\frac{d^2}{dt^2} \phi_t(x) = 0 \implies \exists c \in \mathbb{R}^d : \partial_t \phi_t(x) = c,$$

so using the boundary condition $\phi_t(x) = x$ we get

$$\phi_t(x) = x + tc.$$

This further implies

$$c = \phi_1(x) - x,$$

and doing some algebra we get

$$\phi_t(x) = (1 - t)x + t\phi_1(x).$$

Finally, let us show that (1) $\iff$ (3). By definition we have

$$\partial_t \phi_t(x) = v_t(\phi_t(x)),$$

and, therefore, by the chain rule

$$\begin{aligned} \frac{d^2}{dt^2} \phi_t(x) &= \partial_t v_t(\phi_t(x)) \\ &= \partial_t v_t(\phi_t(x)) + (v_t(x) \cdot \nabla) v_t(\phi_t(x)) \\ &= D_t v_t(\phi_t(x)). \end{aligned}$$

Since $\phi_t : x \mapsto \phi_t(x)$ is a homeomorphism, it is in particular bijective, so we have

$$\frac{d^2}{dt^2} \phi_t(x) = 0 \iff D_t v_t(x) = 0,$$

for all $x \in \mathbb{R}^d$.

□

Lemma 20. *The following equation holds for all $(t, x) \in [0, 1] \times \mathbb{R}^d$:*

$$\partial_t(\rho_t v_t) + \nabla \cdot (\rho_t \mathcal{C}_t) = \rho_t a_t.$$

Proof. For a vector valued test function $\Phi \in C_c^\infty(\mathbb{R}^d; \mathbb{R}^d)$, we have

$$\begin{aligned} \int \Phi(x) \cdot \partial_t(\rho_t(x) v_t(x)) dx &= \partial_t \int \Phi(x) \cdot (\rho_t(x) v_t(x)) dx \\ &= \partial_t \int \Phi(x) \cdot v_t(x) d\rho_t(x) \\ &= \partial_t \mathbb{E} \left[\Phi(X_t) \cdot \mathbb{E} \left[\dot{X}_t \mid X_t \right] \right] \\ &= \partial_t \mathbb{E} \left[\Phi(X_t) \cdot \dot{X}_t \right] \\ &= \mathbb{E} \left[\Phi(X_t) \cdot \ddot{X}_t \right] + \mathbb{E} \left[\left(\nabla \Phi(X_t) \dot{X}_t \right) \cdot \dot{X}_t \right] \\ &= \mathbb{E} \left[\Phi(X_t) \cdot a_t(X_t) \right] + \mathbb{E} \left[\nabla \Phi(X_t) : \dot{X}_t \otimes \dot{X}_t \right] \\ &= \mathbb{E} \left[\Phi(X_t) \cdot a_t(X_t) \right] + \mathbb{E} \left[\nabla \Phi(X_t) : \mathcal{C}_t(X_t) \right] \\ &= \int \Phi(x) \cdot (\rho_t(x) a_t(x)) dx + \int \nabla \Phi(x) : (\rho_t(x) \mathcal{C}_t(x)) dx \\ &= \int \Phi(x) \cdot (\rho_t(x) a_t(x)) dx - \int \Phi(x) \cdot [\nabla \cdot (\rho_t(x) \mathcal{C}_t(x))] dx \end{aligned}$$

where we have only used definitions and the integration-by-parts formula. Since the above holds for all $\Phi \in C_c^\infty(\mathbb{R}^d; \mathbb{R}^d)$, we can conclude that

$$\partial_t(\rho_t v_t) + \nabla \cdot (\rho_t \mathcal{C}_t) = \rho_t a_t,$$

concluding the proof. □

Lemma 21. *The following equation holds for all $(t, x) \in [0, 1] \times \mathbb{R}^d$:*

$$\rho_t D_t v_t + \nabla \cdot (\rho_t \Pi_t) = \rho_t a_t.$$

Proof. By the definition of the Reynolds stress tensor we have

$$\mathcal{C}_t(x) = v_t \otimes v_t + \Pi_t(X_t).$$

Now write $\pi_t^{(i)} \in \mathbb{R}^d$ for the i -th row of Π_t and $v_t^i, a_t^i \in \mathbb{R}$ for the i -th components of v_t and a_t , respectively. Plugging the above display into [\(2\)](#) and looking at the i -th component of the resulting vector we obtain

$$\partial_t (\rho_t v_t^i) + \nabla \cdot (\rho_t v_t^i v_t) + \nabla \cdot (\rho_t \pi_t^{(i)}) = \rho_t a_t^i.$$

Now expanding the differential operators we have

$$v_t^i \partial_t \rho_t + \rho_t \partial_t v_t^i + v_t^i \nabla \cdot (\rho_t v_t) + \rho_t v_t \cdot \nabla v_t^i + \nabla \cdot (\rho_t \pi_t^{(i)}) = \rho_t a_t^i.$$

and rearranging

$$v_t^i \left(\partial_t \rho_t + \nabla \cdot (\rho_t v_t) \right) + \rho_t \left(\partial_t v_t^i + v_t \cdot \nabla v_t^i \right) + \nabla \cdot (\rho_t \pi_t^{(i)}) = \rho_t a_t^i.$$

Using the continuity equation (23), the first parenthesis vanishes. Vectorizing the equation we obtain

$$\rho_t (\partial_t v_t + v_t \cdot \nabla v_t) + \nabla \cdot (\rho_t \Pi_t) = \rho_t a_t.$$

Finally, using the definition of the material derivative we conclude. $\square$

Theorem. Suppose that $X_t = (1-t)X_0 + tX_1$ with $(X_0, X_1) \sim \gamma$ and $\gamma \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ with first and second marginals P_0 and P_1 , respectively. Moreover, suppose that for all $t \in [0, 1]$ we have $\mathbb{E} \|X_t\|^2 < \infty$ and $\mathbb{E} \|\Pi_t(X_t)\|^2 < \infty$ and $t \mapsto \Pi_t(x)$ is continuous at $t = 1$ for all $x \in \mathbb{R}^d$. Then, we have

$$\nabla \cdot (\rho_t \Pi_t) = 0$$

only if there exists a measurable map $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ satisfying $T_\# P_0 = P_1$ such that

$$(\text{id} \times T)_\# P_0 = \gamma,$$

where $\text{id} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the identity map. Moreover, the condition is sufficient in $[0, 1)$ provided the Jacobian of the map T is positive semi-definite at any $x \in \mathbb{R}^d$. In particular, this condition holds for a continuously differentiable optimal transport map T satisfying the above conditions on X_t and Π_t .

Proof. Fix $R > 0$ and consider a cut-off function $\eta_R \in C_c^\infty(\mathbb{R}^d)$ such that $\text{supp}(\eta_R) \subseteq B_{2R}(0)$ and $\text{supp}(1 - \eta_R) \subset B_R(0)^c$. Now consider the test function $\Phi_R(x) \in C_c^\infty(\mathbb{R}^d; \mathbb{R}^d)$ given by $\Phi_R(x) = \eta_R(x)x$ and compute

$$\begin{aligned} & \int \Phi_R(x) \cdot (\nabla \cdot (\rho_t \Pi_t)) \, dx = \\ & - \int \nabla \Phi_R(x) : (\rho_t \Pi_t) \, dx = \\ & - \int \nabla \Phi_R(x) : \Pi_t(x) \, d\rho_t(x) = \\ & - \int_{B_R(0)} I_d : \Pi_t(x) \, d\rho_t(x) + \int_{B_R(0)^c} \nabla \Phi_R(x) : \Pi_t(x) \, d\rho_t(x) = \\ & - \int_{B_R(0)} \text{Tr} \Pi_t(x) \, d\rho_t(x) + \int_{B_R(0)^c} \nabla \Phi_R(x) : \Pi_t(x) \, d\rho_t(x) = \end{aligned}$$

Taking the limit $R \rightarrow \infty$ and using Lemma 22 together with dominated convergence we obtain

$$\int_{\mathbb{R}^d} \text{Tr} \Pi_t(x) \, d\rho_t(x) = 0$$

which is equivalent to

$$\sum_{i=1}^d \int_{\mathbb{R}^d} \text{Var} \left(\dot{X}_t^i \mid X_t = x \right) \, d\rho_t(x) = 0.$$

Now since $\text{Var}(\dot{X}_t^i | X_t = x) \geq 0$ we have that

$$\text{Var}(\dot{X}_t^i | X_t = x) = 0 \text{ for } \rho_t\text{-almost every } x \in \mathbb{R}^d.$$

Thus, there is a Borel measurable function $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that for all $t \in [0, 1]$

$$\dot{X}_t = G(X_t) \text{ a.s.}$$

Taking $t = 0$ this reads $Y = X + G(X)$ almost surely, thus after setting $T = \text{id} + G$ we have

$$Y = T(X) \text{ a.s.}$$

This concludes the proof of the “only if” part of the theorem. For the “if” part, we note that the deterministic map

$$T_t(x) := (1 - t)x + tT(x),$$

satisfies

$$X_t = T_t(x) \text{ a.s.}$$

Now observe that for any $x \neq y \in \mathbb{R}^d$ we have

$$(T(t, x) - T(t, y)) \cdot (x - y) = (1 - t)\|x - y\|^2 + t(T(x) - T(y)) \cdot (x - y).$$

By the fundamental theorem of calculus and the fact that the Jacobian of T is positive semi-definite we have

$$(T(x) - T(y)) \cdot (x - y) = \int_0^1 (x - y) \cdot \nabla T(sx + (1 - s)y) \cdot (x - y) ds \geq 0,$$

and thus

$$(T(t, x) - T(t, y)) \cdot (x - y) > 0 \implies T(t, x) \neq T(t, y).$$

In other words, for any $t \in [0, 1)$ the map $T(t, \cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ has an inverse $T_t^{-1} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ onto its image and we can write

$$X = T_t^{-1}(X_t) \text{ a.s.}$$

It follows that

$$\dot{X}_t = Y - X = T(1, X) - X = T(1, T_t^{-1}(X_t)) - T(t, T_t^{-1}(X_t)) \text{ a.s.}$$

is a measurable function of X_t for all $t \in [0, 1)$, implying that

$$\Pi_t(X_t) = \text{Var}(\dot{X}_t | X_t) = 0 \text{ a.s. for all } t \in [0, 1).$$

By the continuity of $t \mapsto \Pi_t(x)$ at $t = 1$ for all $x \in \mathbb{R}^d$, the above display can, in fact, be extended to hold in the closed interval $t \in [0, 1]$. In other words, we have

$$\Pi_t(x) = 0 \text{ for } \rho_t\text{-a.e. } x \in \mathbb{R}^d \text{ for all } t \in [0, 1],$$

Therefore, we have

$$\nabla \cdot (\rho_t \Pi_t) = 0 \text{ for all } t \in [0, 1].$$

Finally, note that for the optimal transport map T with the squared Euclidean cost between absolutely continuous distributions there is a unique convex potential $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that $T = \nabla \phi$, see [FG21, Theorem 2.5.10]. Thus, the Jacobian of T will be positive semi-definite at any $x \in \mathbb{R}^d$ and the above condition will hold. $\square$

Lemma 22. *Assume that*

$$\begin{aligned}\mathbb{E} \|X_t\|^2 &< \infty \text{ and} \\ \mathbb{E} \|\Pi_t(X_t)\|^2 &< \infty,\end{aligned}$$

for all $t \in [0, 1]$. Then, for all $t, x \in [0, 1] \times \mathbb{R}^d$ there is a collection of random variables $Y_t : \Omega \rightarrow \mathbb{R}$ such that

$$Y_t \in L^1,$$

and

$$\left| \nabla \Phi_R(X_t) : \Pi_t(X_t) \right| \leq Y_t,$$

uniformly in $R > 0$.

Proof. Start by noticing that the cut-off function η_R can be chosen such that

$$\|\eta_R\|_{W^{1,\infty}} \leq C$$

for some constant $C > 0$ independent of R and so for all $x \in \mathbb{R}^d$

$$|\nabla \Phi_R(x)| \lesssim \|x\|.$$

Thus, for any $x \in \mathbb{R}^d$ we can write

$$\begin{aligned}\left| \nabla \Phi_R(x) : \Pi_t(x) \right| &\leq \sum_{ij} \left| \nabla \Phi_R(x) \right| \left| \Pi_t^{ij}(x) \right| \\ &\lesssim \|x\| \sum_{ij} \left| \Pi_t^{ij}(x) \right| \\ &\lesssim \|x\|^2 + \left(\sum_{ij} \left| \Pi_t^{ij}(x) \right| \right)^2 \\ &\lesssim \|x\|^2 + \sum_{ij} \left| \Pi_t^{ij}(x) \right|^2\end{aligned}$$

where in the third line we used the Cauchy-Schwartz inequality and in the third line we used Jensen's inequality, suppressing constants depending on the dimension d . Now we can set

$$Y_t = \|X_t\|^2 + \sum_{ij} \left| \Pi_t^{ij}(X_t) \right|^2$$

which is clearly in L^1 by the assumptions in the statement of the lemma. □

Lemma 23 (Continuity Equation). *We have the identity*

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0$$

for all $t \in [0, 1]$.

Proof. Fix a test function $\varphi \in C_c^\infty(\mathbb{R}^d)$ and compute

$$\begin{aligned}
\int_{\mathbb{R}^d} \varphi(x) \partial_t \rho_t(x) dx &= \partial_t \int_{\mathbb{R}^d} \varphi(x) d\rho_t(x) \\
&= \partial_t \mathbb{E}[\varphi(X_t)] \\
&= \mathbb{E}[\nabla \varphi(X_t) \cdot \dot{X}_t] \\
&= \mathbb{E}[\nabla \varphi(X_t) \cdot v_t(X_t)] \\
&= \int_{\mathbb{R}^d} \nabla \varphi(x) \cdot v_t(x) d\rho_t(x) \\
&= - \int_{\mathbb{R}^d} \varphi(x) \nabla \cdot (\rho_t(x) v_t(x)) dx
\end{aligned}$$

where we used integration-by-parts, the fact that φ has compact support as well as properties of the conditional expectation. Since the above holds for all $\varphi \in C_c^\infty(\mathbb{R}^d)$ we get the desired result. $\square$

B Section 3 proofs

Lemma 24. Recall that a stochastic process $X_\bullet : t \mapsto X_t$ with $X_t \in L^2(\Omega; \mathbb{R}^d)$ for all $t \in [0, 1]$ is said to be mean-square differentiable if there exists a process $\dot{X}_\bullet : t \mapsto \dot{X}_t$ with $\dot{X}_t \in L^2(\Omega; \mathbb{R}^d)$ for all $t \in [0, 1]$ such that

$$\lim_{h \rightarrow 0} \mathbb{E} \left[\left\| \frac{X_{t+h} - X_t}{h} - \dot{X}_t \right\|^2 \right] = 0,$$

for all $t \in [0, 1]$. For any such process, the covariance matrix

$$\Sigma_t := \text{Cov}(X_t, X_t),$$

is differentiable in a mean-square sense and satisfies

$$\dot{\Sigma}_t = \text{Cov}(\dot{X}_t, X_t) + \text{Cov}(X_t, \dot{X}_t).$$

Proof. First, recall that

$$\text{Cov}(X_t, X_t) = \mathbb{E}[X_t X_t^\top] - \mathbb{E}[X_t] \mathbb{E}[X_t]^\top.$$

Examining the two terms separately, we start by defining

$$\Delta_h X_t := \frac{X_{t+h} - X_t}{h}$$

for $h \neq 0$ sufficiently small and recalling that by the mean-square differentiability assumption we have

$$\Delta_h X_t \xrightarrow{L^2} \dot{X}_t \text{ as } h \rightarrow 0.$$

Now by the Cauchy-Schwartz inequality this implies

$$\Delta_h X_t \rightarrow \dot{X}_t \text{ in } L^1 \text{ as } h \rightarrow 0,$$

and thus

$$\frac{d}{dt} (\mathbb{E}[X_t] \mathbb{E}[X_t]^\top) = \mathbb{E}[\dot{X}_t] \mathbb{E}[X_t]^\top + \mathbb{E}[X_t] \mathbb{E}[\dot{X}_t]^\top.$$

Now for the second term we have

$$\begin{aligned} \Delta_h (\mathbb{E}[X_t X_t^\top]) &= \mathbb{E} [\Delta_h (X_t X_t^\top)] \\ &= \frac{1}{h} \mathbb{E} [X_{t+h} X_{t+h}^\top - X_t X_t^\top] \\ &= \frac{1}{h} \mathbb{E} [X_{t+h} (X_{t+h} - X_t)^\top + (X_{t+h} - X_t) X_t^\top] \\ &= \mathbb{E} [X_{t+h} (\Delta_h X_t)^\top + (\Delta_h X_t) X_t^\top], \end{aligned}$$

and using the Cauchy–Schwartz inequality once more we have that

$$\begin{aligned} (\Delta_h X_t) X_t^\top &\xrightarrow{L^1} \dot{X}_t X_t^\top \text{ as } h \rightarrow 0, \\ X_{t+h} (\Delta_h X_t)^\top &\xrightarrow{L^1} X_t \dot{X}_t^\top \text{ as } h \rightarrow 0, \end{aligned}$$

and in the second line we have used that a mean-square differentiable process is also L^2 -continuous. Indeed, one has

$$\begin{aligned} \mathbb{E} [\|X_{t+h} - X_t\|^2] &= h^2 \mathbb{E} \left[\left\| \frac{X_{t+h} - X_t}{h} - \dot{X}_t + \dot{X}_t \right\|^2 \right] \\ &\leq 2 h^2 \mathbb{E} \left[\left\| \frac{X_{t+h} - X_t}{h} - \dot{X}_t \right\|^2 \right] + 2 h^2 \mathbb{E} [\|\dot{X}_t\|^2] \end{aligned}$$

and the right-hand side vanishes as $h \rightarrow 0$. Therefore, we obtain

$$\frac{d}{dt} (\mathbb{E}[X_t X_t^\top]) = \mathbb{E} [X_t \dot{X}_t^\top + \dot{X}_t X_t^\top].$$

Combining the two terms above we conclude that

$$\begin{aligned} \dot{\Sigma}_t &= \frac{d}{dt} (\mathbb{E}[X_t X_t^\top] - \mathbb{E}[X_t] \mathbb{E}[X_t]^\top) \\ &= \mathbb{E} [X_t \dot{X}_t^\top + \dot{X}_t X_t^\top] - \mathbb{E}[\dot{X}_t] \mathbb{E}[X_t]^\top - \mathbb{E}[X_t] \mathbb{E}[\dot{X}_t]^\top \\ &= \text{Cov}(\dot{X}_t, X_t) + \text{Cov}(X_t, \dot{X}_t), \end{aligned}$$

completing the proof. $\square$

C Auxiliary results for Generalized Interpolants

Lemma 25. *Consider any generalized interpolant (F, Q) inducing the process $X_\bullet \in \mathcal{S}(P_0, P)$ and let v be the associated velocity field. Then, the pair (ρ, v) consisting of the time-marginal densities ρ_t of X_t and the velocity field v_t satisfies the continuity equation*

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0,$$

in the distributional sense.

Proof. The proof follows [ABVE23]. For each $t \in [0, 1]$, take any test function $\varphi \in C_c^\infty(\mathbb{R}^d)$ and compute

$$\begin{aligned} \frac{d}{dt} \mathbb{E}[\varphi(X_t)] &= \frac{d}{dt} \int_{\mathbb{R}^d} \varphi(x) \rho_t(x) dx \\ &= \int_{\mathbb{R}^d} \varphi(x) \partial_t \rho_t(x) dx. \end{aligned}$$

On the other hand, by the definition of the velocity field v_t we have

$$\begin{aligned} \frac{d}{dt} \mathbb{E}[\varphi(X_t)] &= \mathbb{E} \left[\nabla \varphi(X_t) \cdot \dot{X}_t \right] \\ &= \mathbb{E} \left[\nabla \varphi(X_t) \cdot \mathbb{E}[\dot{X}_t | X_t] \right] \\ &= \mathbb{E} [\nabla \varphi(X_t) \cdot v(t, X_t)] \\ &= \int_{\mathbb{R}^d} \nabla \varphi(x) \cdot v(t, x) \rho_t(x) dx. \end{aligned}$$

Integrating the last line by parts we obtain

$$\frac{d}{dt} \mathbb{E}[\varphi(X_t)] = - \int_{\mathbb{R}^d} \varphi(x) \nabla \cdot (\rho_t(x) v(t, x)) dx.$$

Combining the above displays yields the desired result. $\square$

Corollary 26. *Consider any $X_\bullet \in \mathcal{S}(P_0, P_1)$ and let v and ϕ be the associated velocity field and flow map, respectively, see Definition [4] for details. Then, we have the distributional equality*

$$\phi(t, X_0) \stackrel{d}{=} X_t \quad \text{for all } t \in [0, 1].$$

In particular, we have $X_1 \stackrel{d}{=} \phi(1, X_0)$ and so

$$P_1 = \phi(1, \cdot)_\# P_0.$$

Proof. If ρ_t is the marginal density of X_t then, by Lemma [25] the pair (ρ, v) must satisfy the continuity equation. On the other hand, let

$$Y_t = \phi(t, X_0).$$

and denote by p_t the marginal density of Y_t . We claim that (p, v) also satisfies the continuity equation. To see this, fix any test function $\varphi \in C_c^\infty(\mathbb{R}^d)$ and compute

$$\begin{aligned} \frac{d}{dt} \mathbb{E}[\varphi(Y_t)] &= \frac{d}{dt} \int_{\mathbb{R}^d} \varphi(x) p_t(x) dx \\ &= \int_{\mathbb{R}^d} \varphi(x) \partial_t p_t(x) dx. \end{aligned}$$

On the other hand, by the definition of the flow map ϕ we have

$$\begin{aligned}
\frac{d}{dt}\mathbb{E}[\varphi(Y_t)] &= \mathbb{E}\left[\nabla\varphi(Y_t) \cdot \frac{d}{dt}\phi(t, X_0)\right] \\
&= \mathbb{E}[\nabla\varphi(Y_t) \cdot v(t, \phi(t, X_0))] \\
&= \mathbb{E}[\nabla\varphi(Y_t) \cdot v(t, Y_t)] \\
&= \int_{\mathbb{R}^d} \nabla\varphi(x) \cdot v(t, x) p_t(x) dx \\
&= - \int_{\mathbb{R}^d} \varphi(x) \nabla \cdot (p_t(x) v(t, x)) dx,
\end{aligned}$$

and in the last line we have integrated by parts. It follows that (p, v) satisfies the continuity equation.

Finally, since

$$p_0 = \rho_0 = \frac{dP_0}{dx}$$

it follows by uniqueness of solutions to the continuity equation that $\rho_t = p_t$ for all $t \in [0, 1]$, completing the proof. $\square$